

Model Analysis

Table of contents

1	Data analysis /Dataframe import and filters	2
1.1	DataFrame import and filters	2
1.2	Metadata	5
2	Data correlation and clustering	6
2.1	Clustermat and cluster list	6
2.2	Features selection	6
3	Model(s) analysis	7
3.1	Metadata	7
3.2	Explanation legend tables	8
3.3	Repartition of train and test	8
3.3.1	External Split: X_train vs X_test	8
3.3.2	Internal CV Folds	9
3.4	Model : Linear	11
3.4.1	Gridsearch	11
3.4.2	Scatter	13
3.4.3	Scatter residuals	14
3.4.4	Learning curve	14
3.5	Model : LGBM	15
3.5.1	Gridsearch	15
3.5.2	Scatter	21
3.5.3	Scatter residuals	22
3.5.4	Learning curve	22
3.6	Model : CatBoost	23
3.6.1	Gridsearch	23
3.6.2	Scatter	29
3.6.3	Scatter residuals	30
3.6.4	Learning curve	30
3.7	Sumup-table	31

4	Plot	31
4.1	Standalone plots	31
4.1.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	31
4.1.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	34
4.1.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	37
4.1.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	40
4.1.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	43
4.1.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	46
4.1.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	49
4.1.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	52
4.2	Compare plots	55
4.2.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	55
4.2.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	57
4.2.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	58
4.2.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	59
4.2.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	61
4.2.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	62
4.2.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	63
4.2.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	65

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	7224
Nombre de lignes df clean	7224
Nombre de features	32
% de NaN	0.0
Taille dataset entraînement	5056
Taille dataset test	2167
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_24h_fft_f1_cph, r_NH3_fft_24h_fft_period1_h, r_NH3_fft_24h_fft_power1, r_NH3_fft_24h_fft_f2_cph, r_NH3_fft_24h_fft_period2_h, r_NH3_fft_24h_fft_power2, r_NH3_fft_24h_fft_f3_cph, r_NH3_fft_24h_fft_period3_h, r_NH3_fft_24h_fft_power3, r_temperature_fft_24h_fft_f1_cph, r_temperature_fft_24h_fft_period1_h, r_temperature_fft_24h_fft_power1, r_temperature_fft_24h_fft_f2_cph, r_temperature_fft_24h_fft_period2_h, r_temperature_fft_24h_fft_power2, r_temperature_fft_24h_fft_f3_cph, r_temperature_fft_24h_fft_period3_h, r_temperature_fft_24h_fft_power3, r_umidity_fft_24h_fft_f1_cph, r_umidity_fft_24h_fft_period1_h, r_umidity_fft_24h_fft_power1, r_umidity_fft_24h_fft_f2_cph, r_umidity_fft_24h_fft_period2_h, r_umidity_fft_24h_fft_power2, r_umidity_fft_24h_fft_f3_cph, r_umidity_fft_24h_fft_period3_h, r_umidity_fft_24h_fft_power3

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

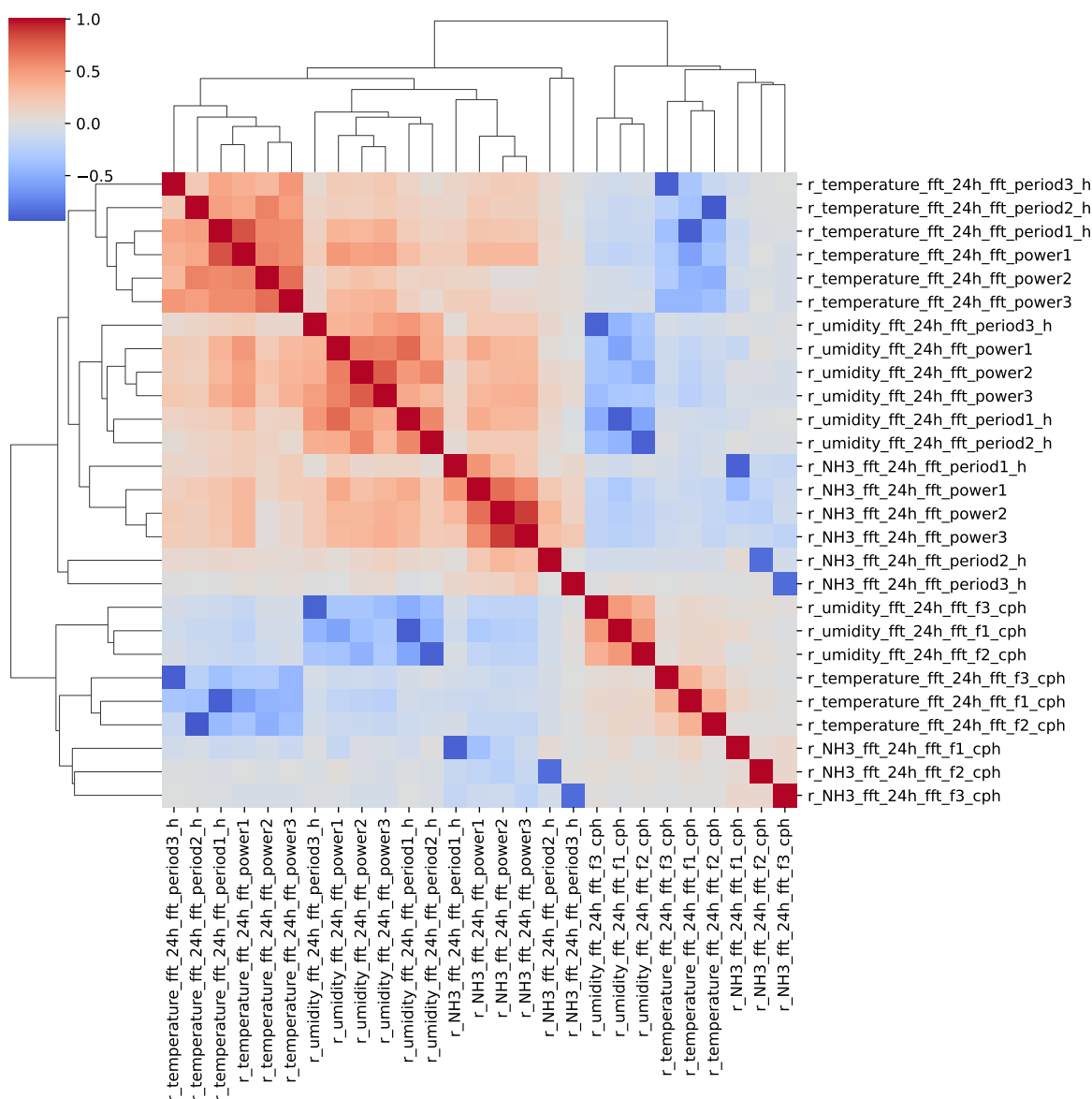
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, , T, H, I ; **cluster 6:** r_NH3_fft_24h_fft_f1_cph, r_NH3_fft_24h_fft_period1_h ; **cluster 7:** r_NH3_fft_24h_fft_power1 ; **cluster 8:** r_NH3_fft_24h_fft_f2_cph, r_NH3_fft_24h_fft_period2_h ; **cluster 9:** r_NH3_fft_24h_fft_power3, r_NH3_fft_24h_fft_power2 ; **cluster 10:** r_NH3_fft_24h_fft_f3_cph, r_NH3_fft_24h_fft_period3_h ; **cluster 11:** r_temperature_fft_24h_fft_f1_cph, r_temperature_fft_24h_fft_period1_h ; **cluster 12:** r_temperature_fft_24h_fft_power1 ; **cluster 13:** r_temperature_fft_24h_fft_period2_h, r_temperature_fft_24h_fft_f2_cph ; **cluster 14:** r_temperature_fft_24h_fft_power2 ; **cluster 15:** r_temperature_fft_24h_fft_f3_cph, r_temperature_fft_24h_fft_period3_h ; **cluster 16:** r_temperature_fft_24h_fft_power3 ; **cluster 17:** r_umidity_fft_24h_fft_f1_cph, r_umidity_fft_24h_fft_period1_h ; **cluster 18:** r_umidity_fft_24h_fft_power1 ; **cluster 19:** r_umidity_fft_24h_fft_f2_cph, r_umidity_fft_24h_fft_period2_h ; **cluster 20:** r_umidity_fft_24h_fft_power2 ; **cluster 21:** r_umidity_fft_24h_fft_f3_cph, r_umidity_fft_24h_fft_period3_h ; **cluster 22:** r_umidity_fft_24h_fft_power3 ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Chooosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_24h_fft_f1_cph, r_NH3_fft_24h_fft_power1, r_NH3_fft_24h_fft_f2_cph, r_NH3_fft_24h_fft_power3, r_NH3_fft_24h_fft_f3_cph, r_temperature_fft_24h_fft_f1_cph, r_temperature_fft_24h_fft_power1, r_temperature_fft_24h_fft_period2_h, r_temperature_fft_24h_fft_power2, r_temperature_fft_24h_fft_f3_cph, r_temperature_fft_24h_fft_power3, r_umidity_fft_24h_fft_f1_cph, r_umidity_fft_24h_fft_power1, r_umidity_fft_24h_fft_f2_cph, r_umidity_fft_24h_fft_power2, r_umidity_fft_24h_fft_f3_cph, r_umidity_fft_24h_fft_power3



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_24h_fft_f1_cph, r_NH3_fft_24h_fft_power1, r_NH3_fft_24h_fft_f2_cph, r_NH3_fft_24h_fft_power3, r_NH3_fft_24h_fft_f3_cph, r_temperature_fft_24h_fft_f1_cph, r_temperature_fft_24h_fft_power1,

r_temperature_fft_24h_fft_period2_h, r_temperature_fft_24h_fft_power2, r_temperature_fft_24h_fft_f3_cph, r_temperature_fft_24h_fft_power3, r_umidity_fft_24h_fft_f1_cph, r_umidity_fft_24h_fft_power1, r_umidity_fft_24h_fft_f2_cph, r_umidity_fft_24h_fft_power2, r_umidity_fft_24h_fft_f3_cph, r_umidity_fft_24h_fft_power3

Len(Features__X): 22

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

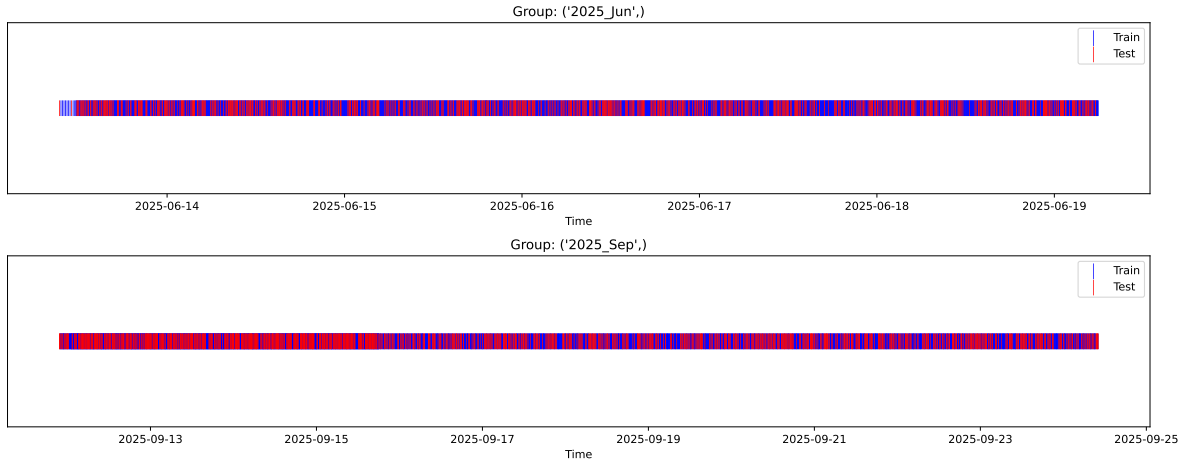
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_{train} vs X_{test}

External Split: Train: 5056 samples Test: 2168 samples



3.3.2 Internal CV Folds

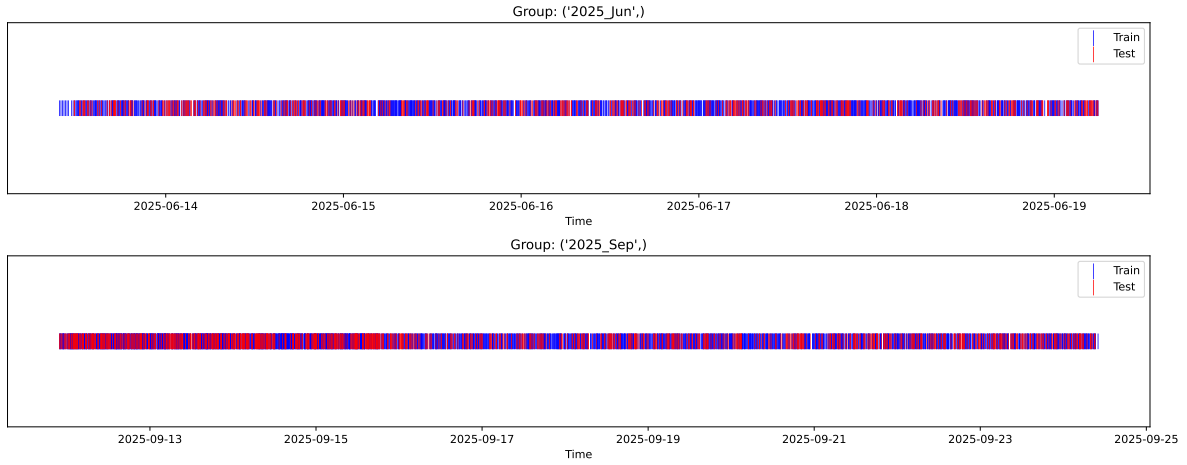
Fold 1/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



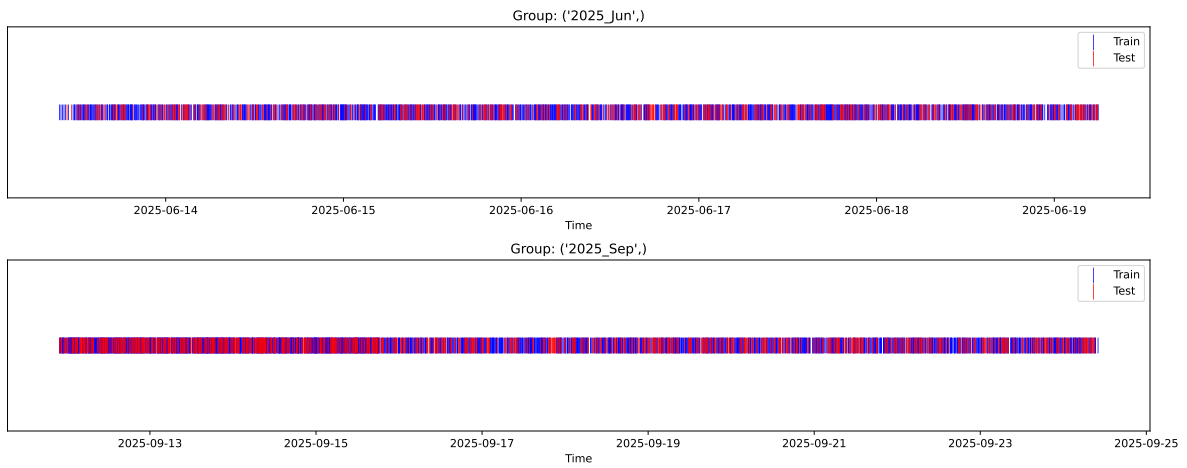
Fold 2/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



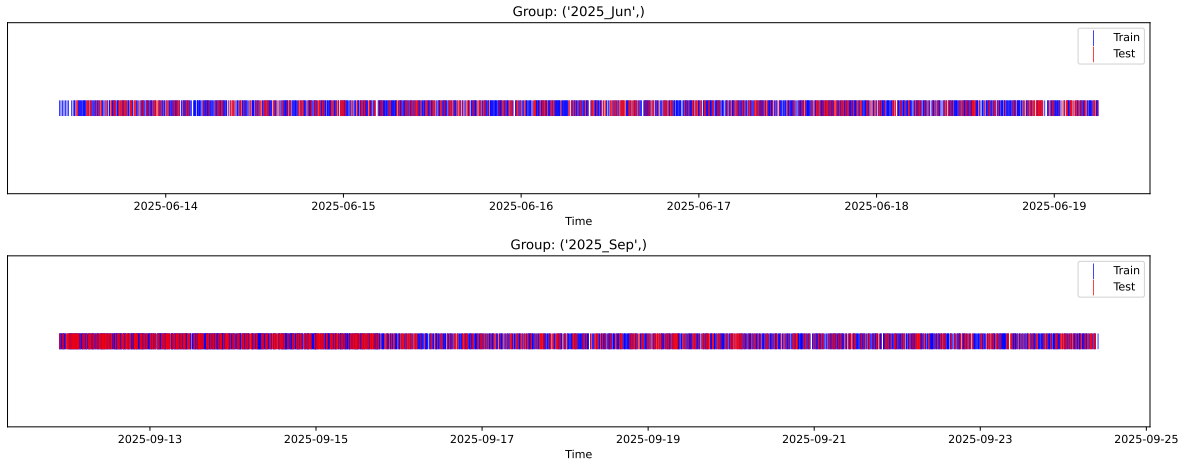
Fold 3/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



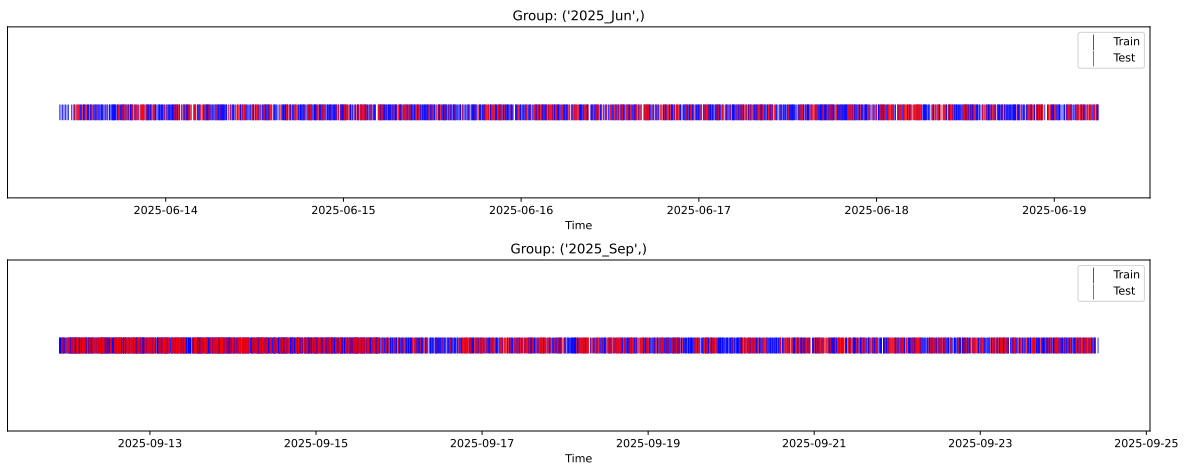
Fold 4/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.346, std=0.787, min=0.959, max=8.190
 y_{test} : mean=2.382, std=0.828, min=1.071, max=13.845

Fold 0 — y_{test} : mean=2.35, std=0.80, y_{train} : mean=2.34, std=0.78

Fold 1 — y_{test} : mean=2.33, std=0.75, y_{train} : mean=2.35, std=0.80

Fold 2 — y_{test} : mean=2.37, std=0.83, y_{train} : mean=2.34, std=0.77

Fold 3 — y_test: mean=2.34, std=0.78, y_train: mean=2.35, std=0.79

Fold 4 — y_test: mean=2.32, std=0.80, y_train: mean=2.36, std=0.78

train scores CV: [0.43308653 0.41695801 0.41827769 0.42245053 0.42150791]

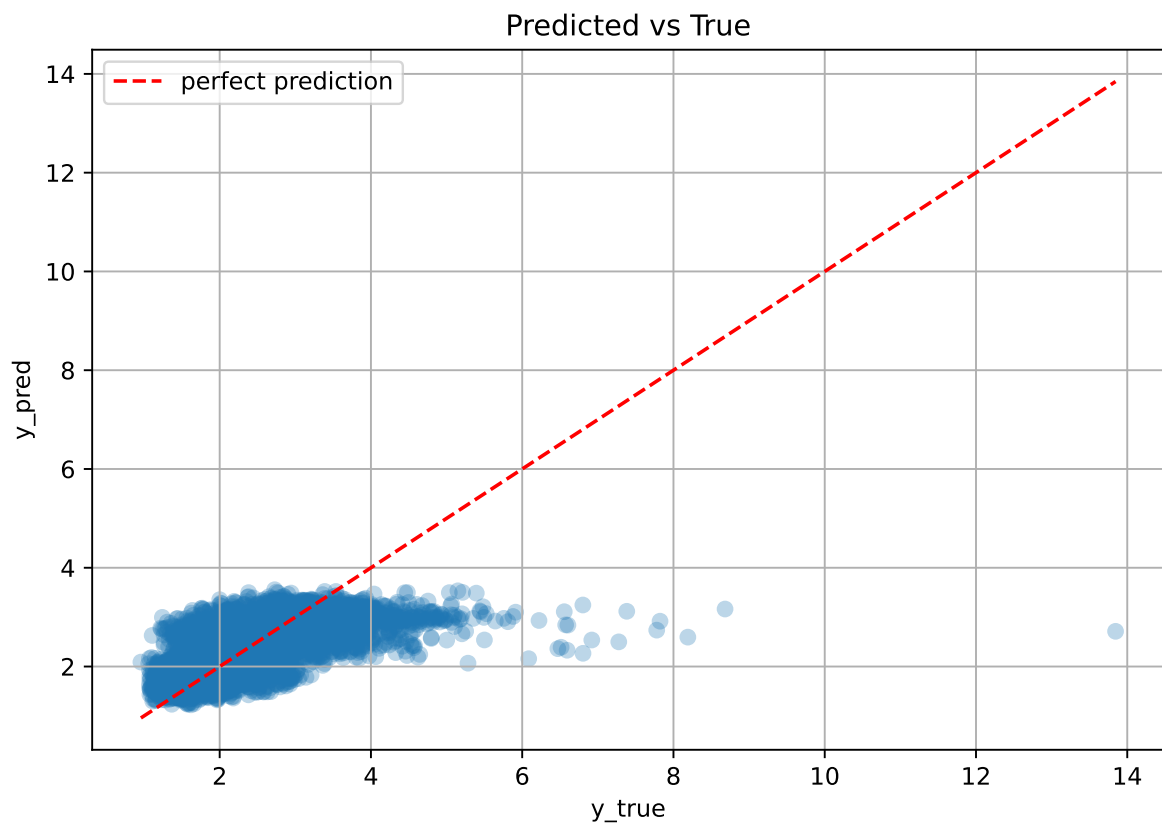
test scores CV: [0.38974851 0.42797038 0.41869479 0.41482522 0.41648885]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4225	0.4135	0.0127	0.3736	0.3754	-0.0382	0.9077	1.0215

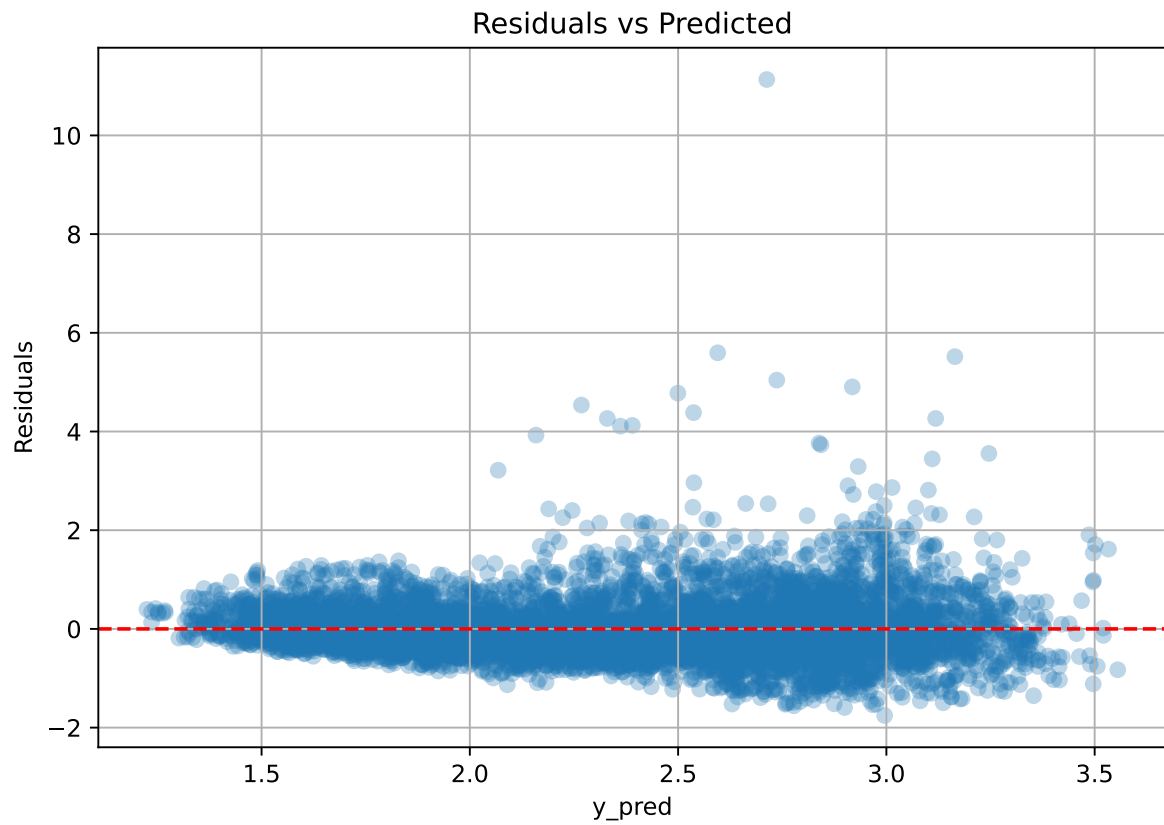
Coefficients:

coef_r_CO2 = -0.0325, coef_r_NH3 = -0.0550, coef_r_NH3_fft_24h_fft_f1_cph = -0.0086, coef_r_NH3_fft_24h_fft_f2_cph = -0.0093, coef_r_NH3_fft_24h_fft_f3_cph = -0.0076, coef_r_NH3_fft_24h_fft_power1 = 0.0286, coef_r_NH3_fft_24h_fft_power3 = 0.1030, coef_r_THI = 0.7694, coef_r_temperature = -0.9280, coef_r_temperature_fft_24h_fft_f1_cph = 0.0789, coef_r_temperature_fft_24h_fft_f3_cph = 0.0325, coef_r_temperature_fft_24h_fft_period2_h = -0.0335, coef_r_temperature_fft_24h_fft_power1 = -0.0009, coef_r_temperature_fft_24h_fft_power2 = 0.0742, coef_r_temperature_fft_24h_fft_power3 = 0.0056, coef_r_umidity = -0.6786, coef_r_umidity_fft_24h_fft_f1_cph = 0.0735, coef_r_umidity_fft_24h_fft_f2_cph = 0.0349, coef_r_umidity_fft_24h_fft_f3_cph = -0.0276, coef_r_umidity_fft_24h_fft_power1 = 0.0637, coef_r_umidity_fft_24h_fft_power2 = -0.0203, coef_r_umidity_fft_24h_fft_power3 = 0.0076, intercept = 2.3462,

3.4.2 Scatter



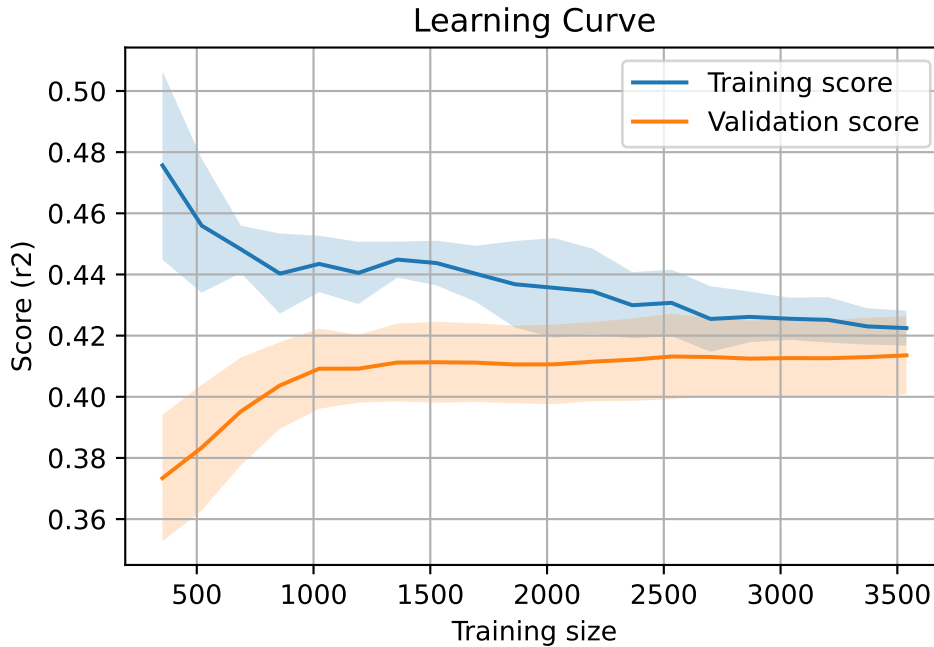
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7224, `Len(training)` : 5057,

`Len(training_real)` : 3540, `Len(test_of_training)` : 1517, `Len(test)` : 2167

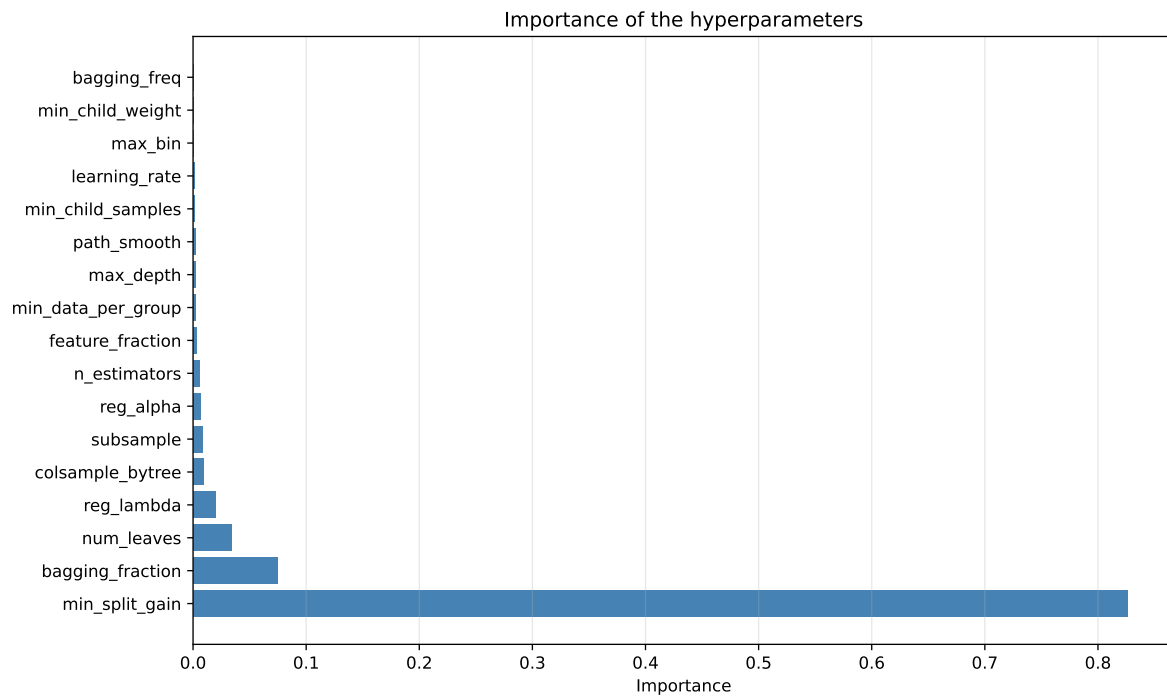
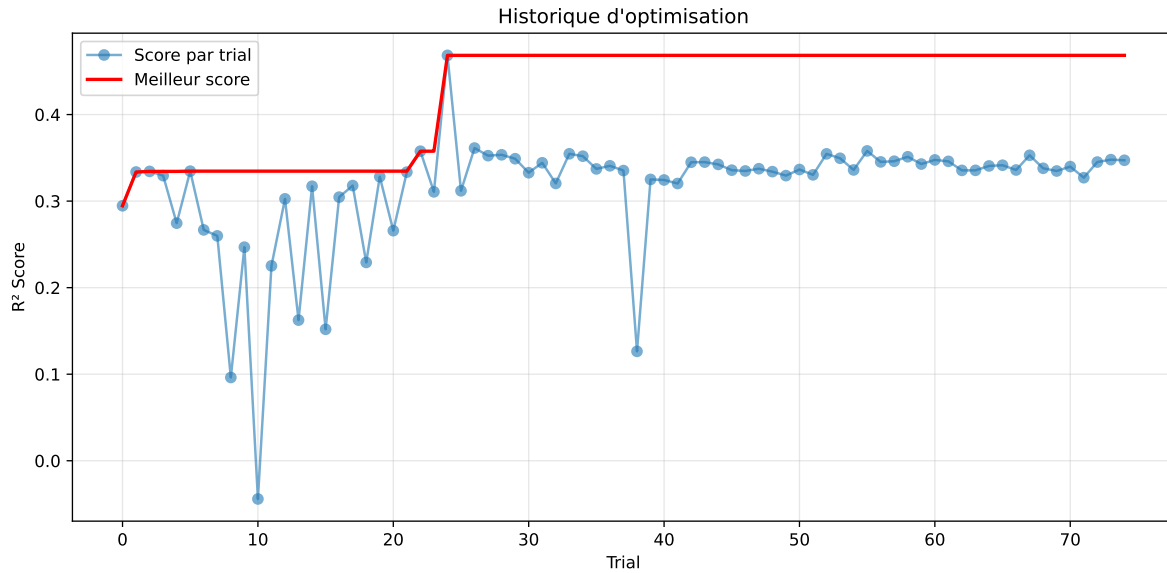


3.5 Model : LGBM

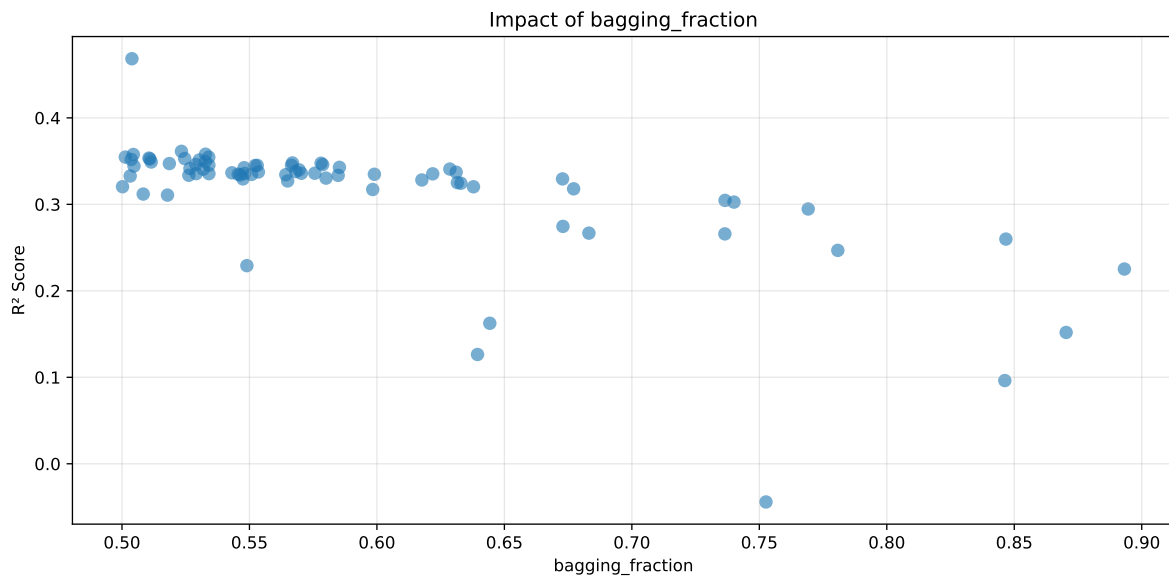
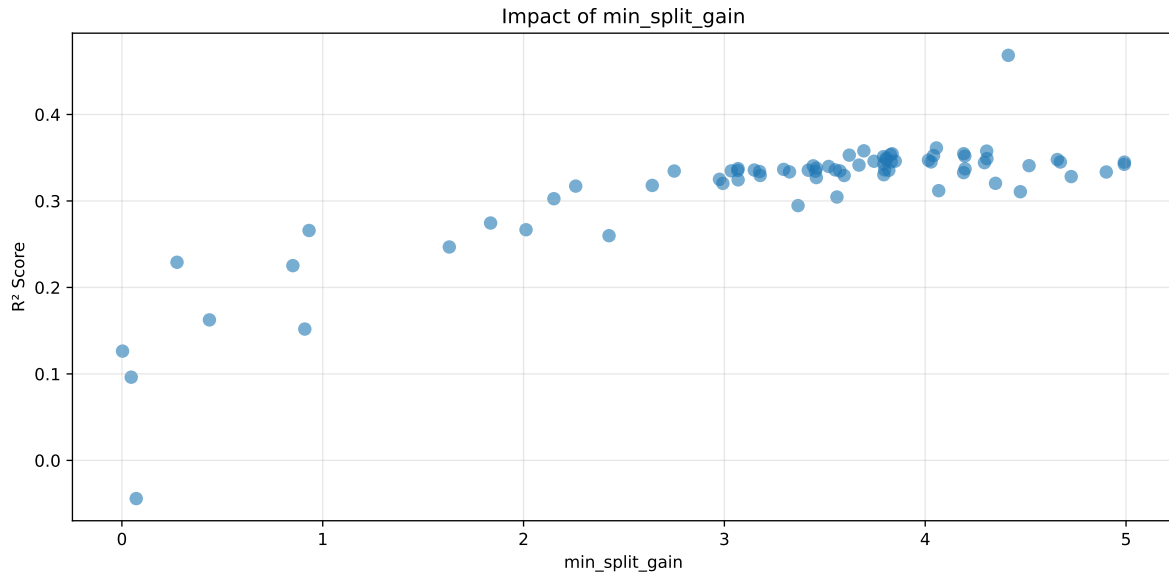
3.5.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.346, std=0.787, min=0.959, max=8.190
 y_{test} : mean=2.382, std=0.828, min=1.071, max=13.845

===== TOP Importance FEATURES =====
 feature importance r_umidity 150 r_NH3_fft_24h_fft_power3 129 r_NH3 105 r_temperature
 67 r_temperature_fft_24h_fft_f1_cph 58 r_THI 50 r_CO2 38 r_NH3_fft_24h_fft_power1
 29 r_umidity_fft_24h_fft_f1_cph 20 r_temperature_fft_24h_fft_power1 10 r_umid-
 ity_fft_24h_fft_power3 3 r_temperature_fft_24h_fft_power2 2 r_temperature_fft_24h_fft_f3_cph
 2 r_umidity_fft_24h_fft_power2 2 r_temperature_fft_24h_fft_power3 1 r_umid-
 ity_fft_24h_fft_f2_cph 1 r_NH3_fft_24h_fft_f3_cph 0 r_NH3_fft_24h_fft_f2_cph
 0 r_temperature_fft_24h_fft_period2_h 0 r_NH3_fft_24h_fft_f1_cph 0 r_umid-
 ity_fft_24h_fft_power1 0 r_umidity_fft_24h_fft_f3_cph 0

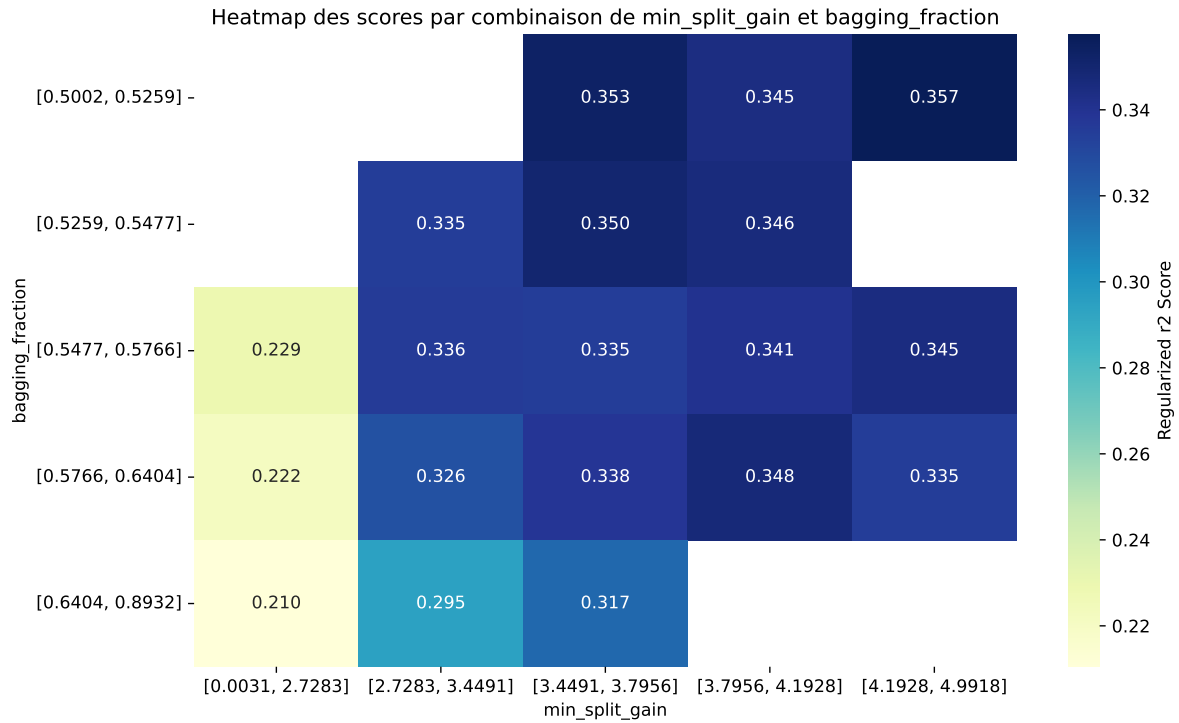


Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4897	0.4684	0.4164	0.0119	0.4684	0.4389	-0.0295	0.937	1.0455
2	0.5177	0.4916	0.4362	0.0127	0.3613	0.4655	-0.0261	0.947	1.053
3	0.5211	0.4939	0.4395	0.0082	0.358	0.4652	-0.0287	0.9418	1.055
4	0.5077	0.4827	0.4303	0.0136	0.3577	0.4604	-0.0224	0.9537	1.0518
5	0.5077	0.4822	0.4275	0.0129	0.3547	0.457	-0.0252	0.9478	1.0529
6	0.5265	0.4979	0.4413	0.0116	0.3546	0.4606	-0.0372	0.9252	1.0576
7	0.526	0.4972	0.4417	0.0075	0.3535	0.4642	-0.033	0.9336	1.0578
8	0.507	0.4813	0.4285	0.0085	0.353	0.461	-0.0202	0.9579	1.0533
9	0.5407	0.5093	0.4531	0.0077	0.3525	0.4762	-0.0331	0.935	1.0616
10	0.5087	0.4826	0.4307	0.0104	0.3519	0.4627	-0.0199	0.9588	1.0542
---	---	---	---	---	---	---	---	---	---
75	0.9088	0.7499	0.6993	0.0099	-0.0442	0.7274	-0.0225	0.97	1.2118

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.021607819910384636, 'reg__num_leaves': 20, 'reg__subsample': 0.744340392892368, 'reg__colsample_bytree': 0.7736977448236492, 'reg__min_child_samples': 22, 'reg__reg_alpha': 9.707477504807215, 'reg__reg_lambda': 49.85364612598128, 'reg__min_split_gain': 4.413724156383154, 'reg__path_smooth': 1.215463327814272, 'reg__min_child_weight':

0.3477417543343749, 'reg__feature_fraction': 0.6963446839503773, 'reg__bagging_fraction': 0.5038984513398881, 'reg__bagging_freq': 3, 'reg__max_bin': 63, 'reg__min_data_per_group': 155}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.023895932758318722, 'reg__num_leaves': 31, 'reg__subsample': 0.7355096359774894, 'reg__colsample_bytree': 0.7180482496778614, 'reg__min_child_samples': 21, 'reg__reg_alpha': 9.602510318952778, 'reg__reg_lambda': 0.4611885807748912, 'reg__min_split_gain': 4.055864057247127, 'reg__path_smooth': 0.9950496302393049, 'reg__min_child_weight': 0.3445931378582889, 'reg__feature_fraction': 0.8125582724625919, 'reg__bagging_fraction': 0.5233311571715368, 'reg__bagging_freq': 3, 'reg__max_bin': 67, 'reg__min_data_per_group': 153}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.029081172134392697, 'reg__num_leaves': 29, 'reg__subsample': 0.7704758954479631, 'reg__colsample_bytree': 0.752753428571782, 'reg__min_child_samples': 17, 'reg__reg_alpha': 9.309917360602892, 'reg__reg_lambda': 5.240164457922942, 'reg__min_split_gain': 3.6944270973737416, 'reg__path_smooth': 0.9238157075211753, 'reg__min_child_weight': 0.1744237642334225, 'reg__feature_fraction': 0.7032314127977797, 'reg__bagging_fraction': 0.5327757982681748, 'reg__bagging_freq': 3, 'reg__max_bin': 79, 'reg__min_data_per_group': 136}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.02065261946191923, 'reg__num_leaves': 16, 'reg__subsample': 0.8873537447518147, 'reg__colsample_bytree': 0.782852546857861, 'reg__min_child_samples': 21, 'reg__reg_alpha': 9.49028777474042, 'reg__reg_lambda': 0.6757769229009654, 'reg__min_split_gain': 4.306166560612156, 'reg__path_smooth': 0.6146658400670578, 'reg__min_child_weight': 0.3864830538907283, 'reg__feature_fraction': 0.7330197958274801, 'reg__bagging_fraction': 0.5044524585583836, 'reg__bagging_freq': 2, 'reg__max_bin': 67, 'reg__min_data_per_group': 158}

Rank 5: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.0253763026862693, 'reg__num_leaves': 21, 'reg__subsample': 0.7481257484405047, 'reg__colsample_bytree': 0.7276438918798945, 'reg__min_child_samples': 23, 'reg__reg_alpha': 9.991326077111703, 'reg__reg_lambda': 9.304383845209818, 'reg__min_split_gain': 4.191695044159175, 'reg__path_smooth': 1.6322871334657454, 'reg__min_child_weight': 0.9318939206400566, 'reg__feature_fraction': 0.8043027471978532, 'reg__bagging_fraction': 0.5013112084470327, 'reg__bagging_freq': 2, 'reg__max_bin': 115, 'reg__min_data_per_group': 172}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.02857235076446774, 'reg__num_leaves': 36, 'reg__subsample': 0.7720028823315451, 'reg__colsample_bytree': 0.7577467523668823, 'reg__min_child_samples': 17, 'reg__reg_alpha': 8.509895220128449, 'reg__reg_lambda': 5.297951791546888, 'reg__min_split_gain': 3.8354857157100573, 'reg__path_smooth': 0.39301848604363987, 'reg__min_child_weight':

0.19173923705799356, 'reg__feature_fraction': 0.8625191858528605, 'reg__bagging_fraction': 0.5340433088342698, 'reg__bagging_freq': 3, 'reg__max_bin': 79, 'reg__min_data_per_group': 162}

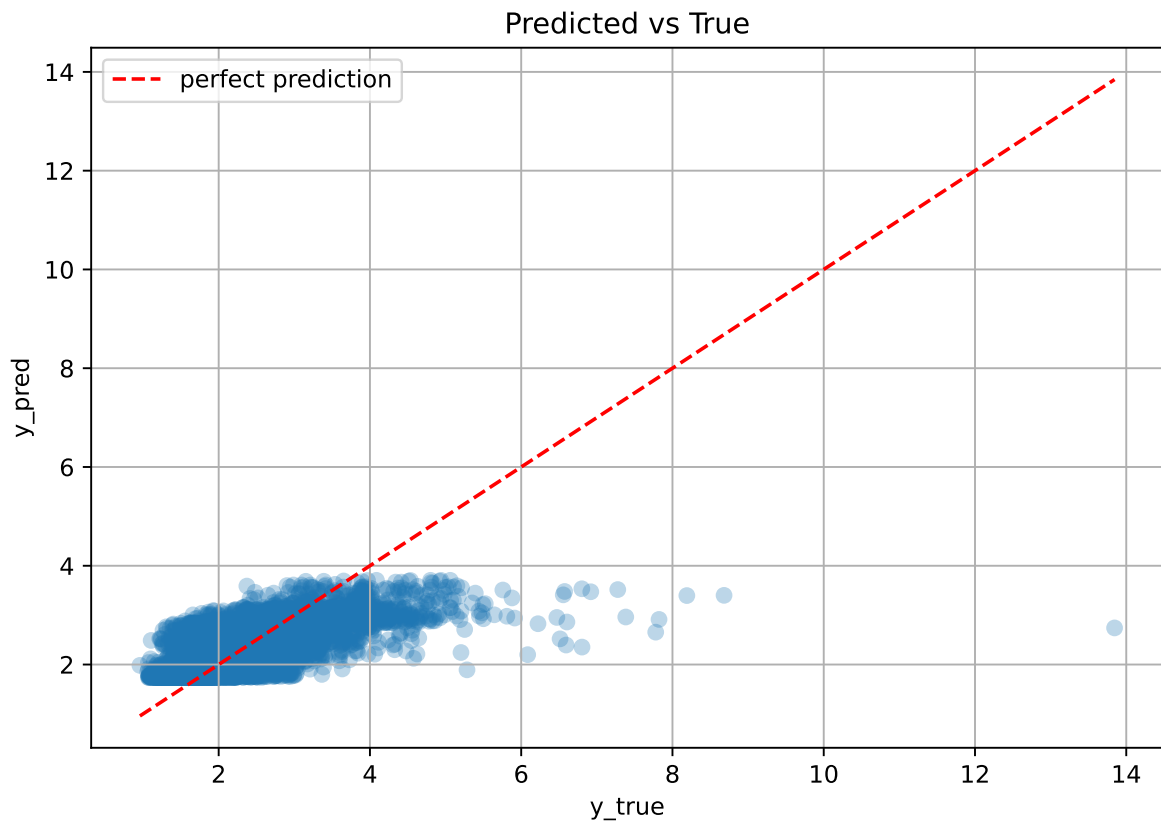
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.02334752549166267, 'reg__num_leaves': 33, 'reg__subsample': 0.7464497586586681, 'reg__colsample_bytree': 0.742664235262988, 'reg__min_child_samples': 21, 'reg__reg_alpha': 8.517368577760578, 'reg__reg_lambda': 0.9825351501087418, 'reg__min_split_gain': 3.8265433450927593, 'reg__path_smooth': 3.3454382706618295, 'reg__min_child_weight': 0.3322258359937656, 'reg__feature_fraction': 0.8162744862690609, 'reg__bagging_fraction': 0.5105808296988856, 'reg__bagging_freq': 3, 'reg__max_bin': 71, 'reg__min_data_per_group': 152}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.02078166537617384, 'reg__num_leaves': 36, 'reg__subsample': 0.708681220211265, 'reg__colsample_bytree': 0.8391138106605075, 'reg__min_child_samples': 12, 'reg__reg_alpha': 9.403871646313254, 'reg__reg_lambda': 33.449563408829086, 'reg__min_split_gain': 3.6219607823721067, 'reg__path_smooth': 1.3493887627445122, 'reg__min_child_weight': 0.0658732565285138, 'reg__feature_fraction': 0.6908433288262793, 'reg__bagging_fraction': 0.524599552558333, 'reg__bagging_freq': 4, 'reg__max_bin': 94, 'reg__min_data_per_group': 184}

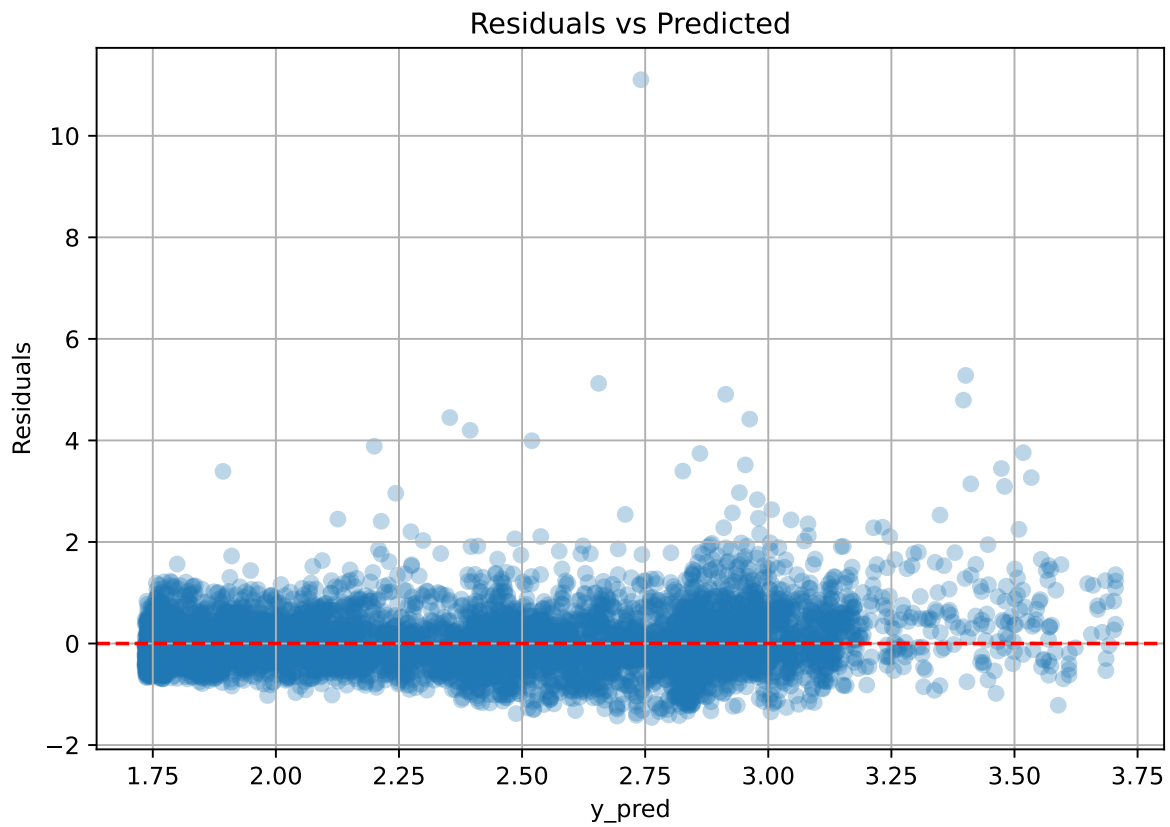
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.02333020553510847, 'reg__num_leaves': 33, 'reg__subsample': 0.7488058801816299, 'reg__colsample_bytree': 0.7238926676874299, 'reg__min_child_samples': 21, 'reg__reg_alpha': 5.008371606753808, 'reg__reg_lambda': 1.4495747514172628, 'reg__min_split_gain': 4.041381806264875, 'reg__path_smooth': 3.3602428921156378, 'reg__min_child_weight': 0.28521341607284123, 'reg__feature_fraction': 0.8244505940464387, 'reg__bagging_fraction': 0.510977539690286, 'reg__bagging_freq': 3, 'reg__max_bin': 67, 'reg__min_data_per_group': 156}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.025411618948225877, 'reg__num_leaves': 20, 'reg__subsample': 0.8427455987504532, 'reg__colsample_bytree': 0.7088023730239769, 'reg__min_child_samples': 24, 'reg__reg_alpha': 9.748296310098016, 'reg__reg_lambda': 9.079662838290288, 'reg__min_split_gain': 4.1971939881710565, 'reg__path_smooth': 0.5970230018854545, 'reg__min_child_weight': 1.4304127987235258, 'reg__feature_fraction': 0.663514253840654, 'reg__bagging_fraction': 0.5036480380644098, 'reg__bagging_freq': 2, 'reg__max_bin': 110, 'reg__min_data_per_group': 175}

3.5.2 Scatter



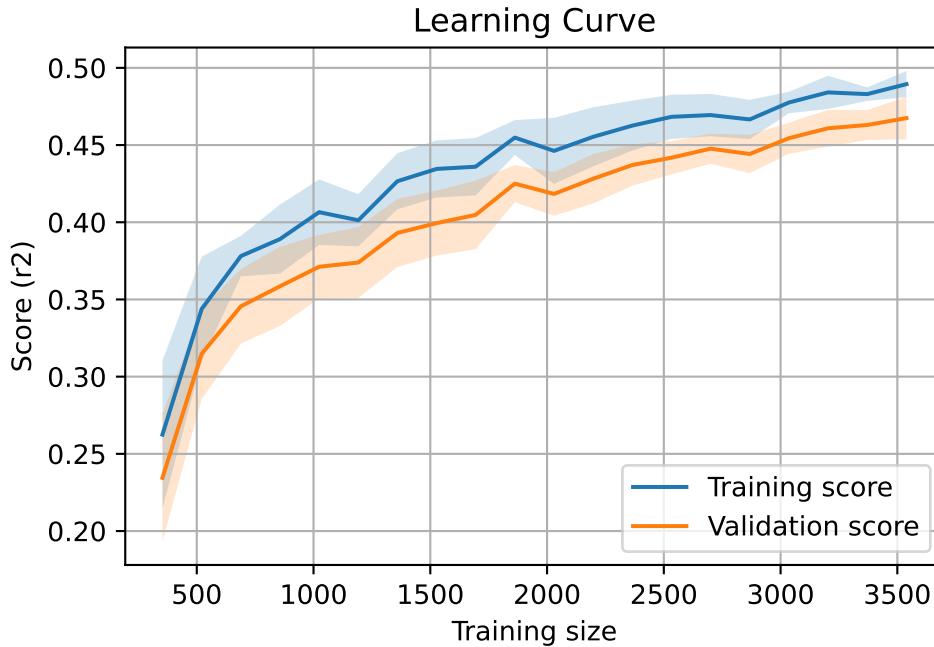
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 7224, **Len(training)** : 5057,

Len(training_real) : 3540, **Len(test_of_training)** : 1517, **Len(test)** : 2167



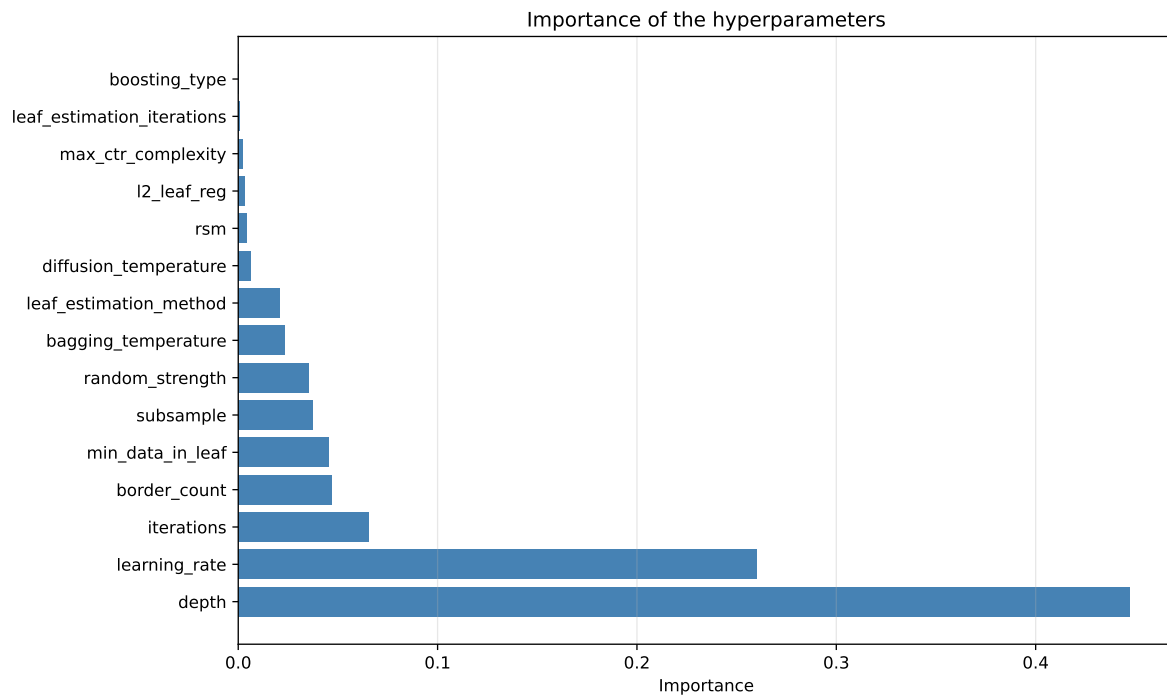
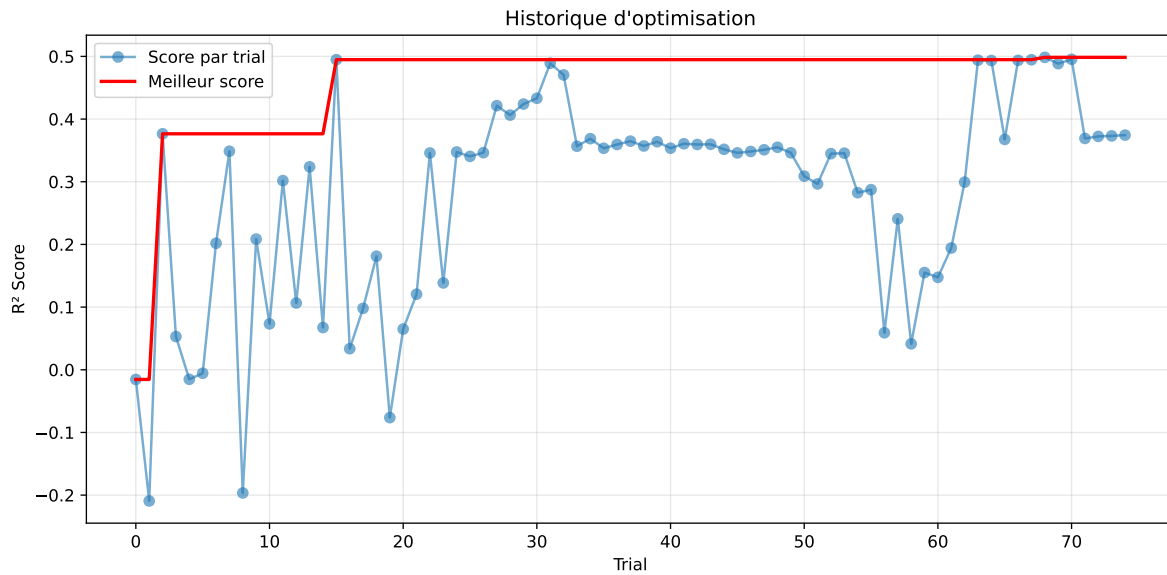
3.6 Model : CatBoost

3.6.1 Gridsearch

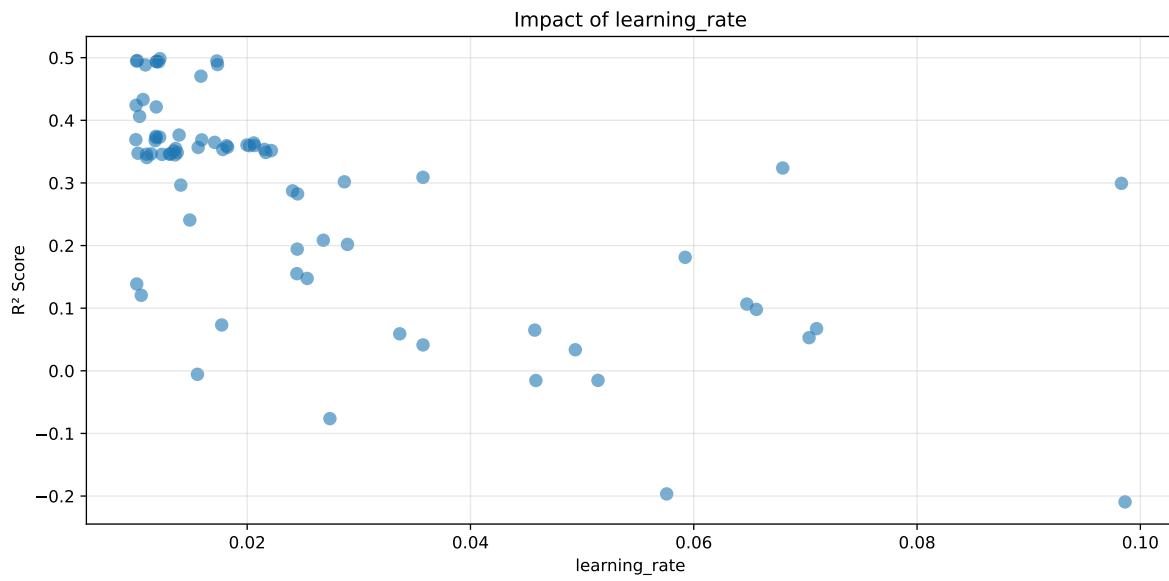
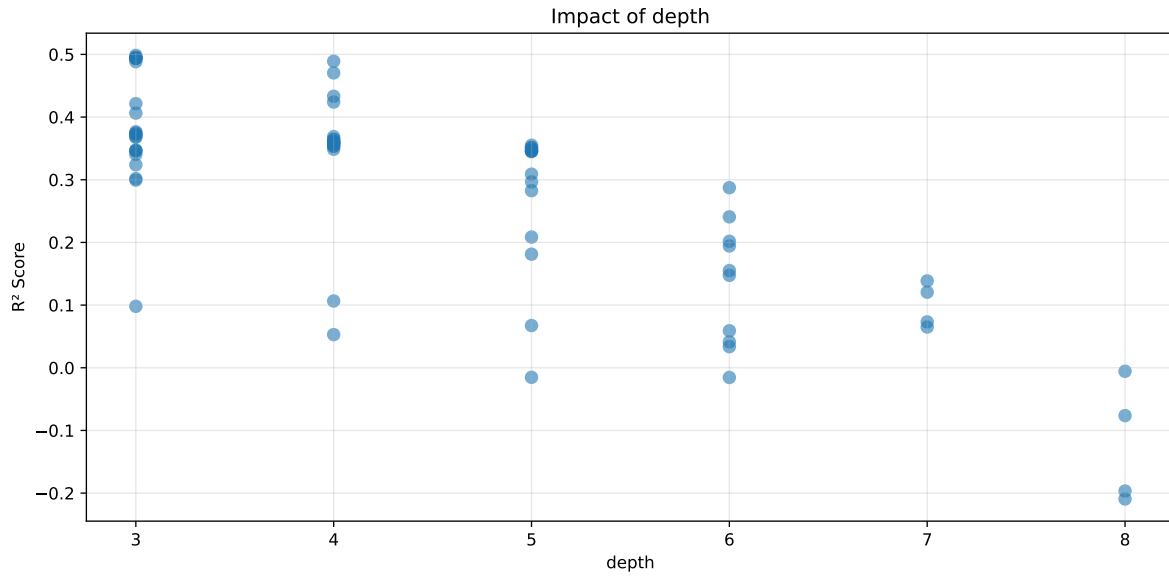
Distribution y_train vs y_test: y_train: mean=2.346, std=0.787, min=0.959, max=8.190
y_test: mean=2.382, std=0.828, min=1.071, max=13.845

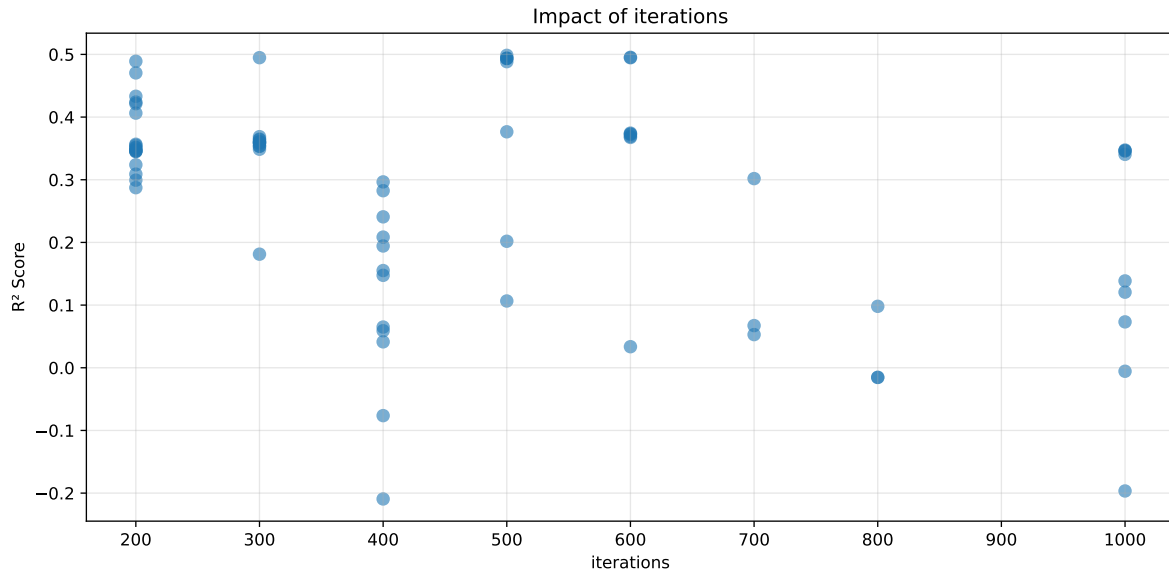
===== TOP Importance FEATURES =====

feature	importance
r_umidity	40.513260
r_temperature	16.143364
r_NH3	12.840972
r_NH3_fft_24h_fft_power3	9.336016
r_THI	4.703315
r_temperature_fft_24h_fft_f1_cph	2.630604
r_NH3_fft_24h_fft_power1	2.453316
r_CO2	1.875951
r_temperature_fft_24h_fft_power1	1.770754
r_umidity_fft_24h_fft_f1_cph	0.939084
r_temperature_fft_24h_fft_power2	0.897538
r_umidity_fft_24h_fft_power1	0.795832
r_umidity_fft_24h_fft_power3	0.708601
r_temperature_fft_24h_fft_period2_h	0.619682
r_umidity_fft_24h_fft_f3_cph	0.566464
r_NH3_fft_24h_fft_f1_cph	0.552403
r_temperature_fft_24h_fft_power3	0.469253
r_umidity_fft_24h_fft_power2	0.467127
r_umidity_fft_24h_fft_f2_cph	0.439160
r_NH3_fft_24h_fft_f3_cph	0.437062
r_NH3_fft_24h_fft_f2_cph	0.435764
r_temperature_fft_24h_fft_f3_cph	0.404478



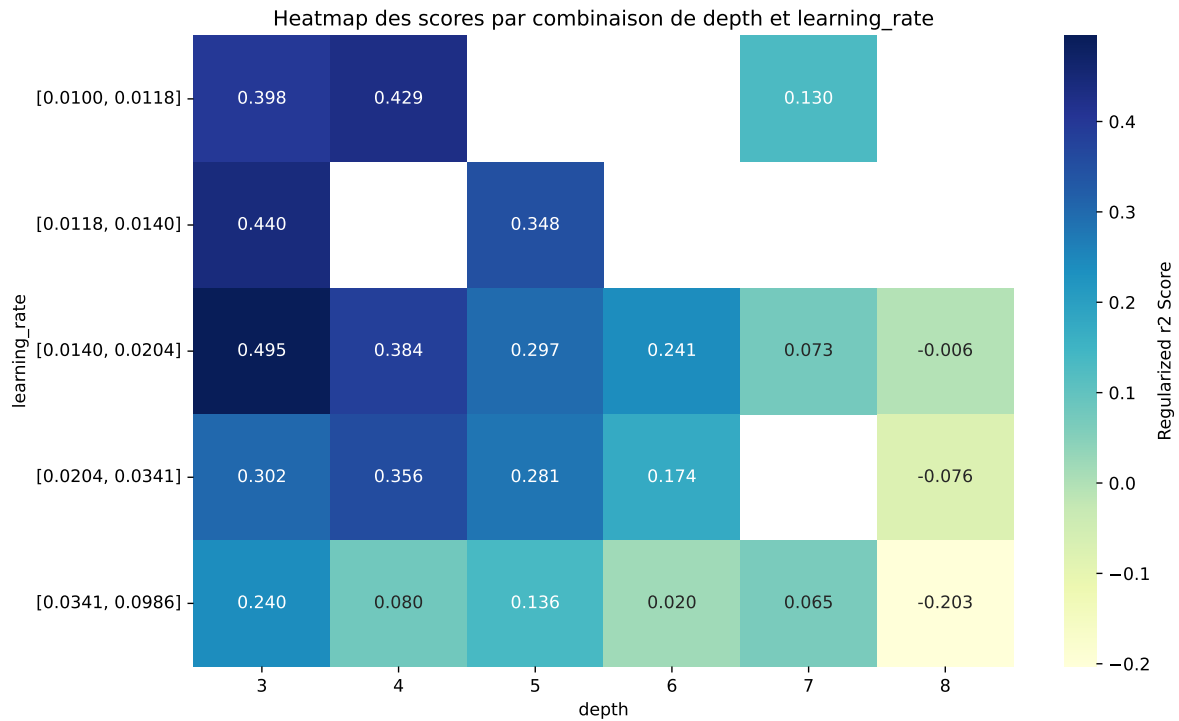
Paramètres avec >5% d'importance : ['depth', 'learning_rate', 'iterations']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5226	0.4985	0.4457	0.0077	0.4985	0.4512	-0.0473	0.9052	1.0484
2	0.519	0.4954	0.4413	0.0068	0.4954	0.4481	-0.0473	0.9046	1.0477
3	0.5186	0.4948	0.4403	0.0085	0.4948	0.4452	-0.0496	0.8997	1.0481
4	0.5179	0.4947	0.4419	0.0085	0.4947	0.4482	-0.0465	0.906	1.0469
5	0.5169	0.494	0.4402	0.0084	0.494	0.4461	-0.0479	0.9031	1.0464
6	0.516	0.4937	0.4414	0.01	0.4937	0.4483	-0.0453	0.9082	1.0453
7	0.516	0.4934	0.4412	0.0089	0.4934	0.4494	-0.044	0.9108	1.0458
8	0.5134	0.4891	0.4356	0.0118	0.4891	0.446	-0.0431	0.9119	1.0497
9	0.5103	0.4884	0.4348	0.0083	0.4884	0.4407	-0.0477	0.9023	1.0449
10	0.4924	0.4705	0.4218	0.0107	0.4705	0.4309	-0.0395	0.916	1.0466
---	---	---	---	---	---	---	---	---	---
75	0.9802	0.7819	0.7266	0.0103	-0.2093	0.7566	-0.0253	0.9676	1.2535

Hyperparameters:

Rank 1: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.012184921313037196, 'reg__l2_leaf_reg': 4.531957486568171, 'reg__bagging_temperature': 1.7901873311617482, 'reg__random_strength': 4.438855143604846, 'reg__subsample': 0.848834940157865, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.7805765652190293, 'reg__border_count': 65, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 4748.159306655302}

Rank 2: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.010158175359872424, 'reg__l2_leaf_reg': 30.726339160533072, 'reg__bagging_temperature': 1.1241884633382506, 'reg__random_strength': 4.410446526556114, 'reg__subsample': 0.8484804162153186, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.7750747722736435, 'reg__border_count': 65, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 4774.801833700428}

Rank 3: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01728497357398152, 'reg__l2_leaf_reg': 17.680079157749372, 'reg__bagging_temperature': 2.6774226152636933, 'reg__random_strength': 3.6841387691155414, 'reg__subsample': 0.8553677737648732, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.8824433503833236, 'reg__border_count': 68, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 9268.923004409642}

Rank 4: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.010088969412400034, 'reg__l2_leaf_reg': 27.605426299566425, 'reg__bagging_temperature': 1.7512180454006792, 'reg__random_strength': 4.49281555184216, 'reg__subsample': 0.8454281180423591, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 10, 'reg__rsm':

0.7891301186697259, 'reg__border_count': 72, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 4641.129669883897}

Rank 5: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.01186669824905242, 'reg__l2_leaf_reg': 27.82957691179604, 'reg__bagging_temperature': 1.707581609364504, 'reg__random_strength': 4.571726732251443, 'reg__subsample': 0.8742168128993343, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.8977115000452068, 'reg__border_count': 74, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4704.155310675319}

Rank 6: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.011895232321695814, 'reg__l2_leaf_reg': 4.205432143470457, 'reg__bagging_temperature': 1.7496494517822334, 'reg__random_strength': 4.656181323585974, 'reg__subsample': 0.845628783420688, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.6747154764680834, 'reg__border_count': 73, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4571.238930164211}

Rank 7: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.012077973891542724, 'reg__l2_leaf_reg': 23.468905409143094, 'reg__bagging_temperature': 1.8192669817551739, 'reg__random_strength': 4.623083775165188, 'reg__subsample': 0.8769380903887041, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.7871018183105073, 'reg__border_count': 70, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4232.946303005177}

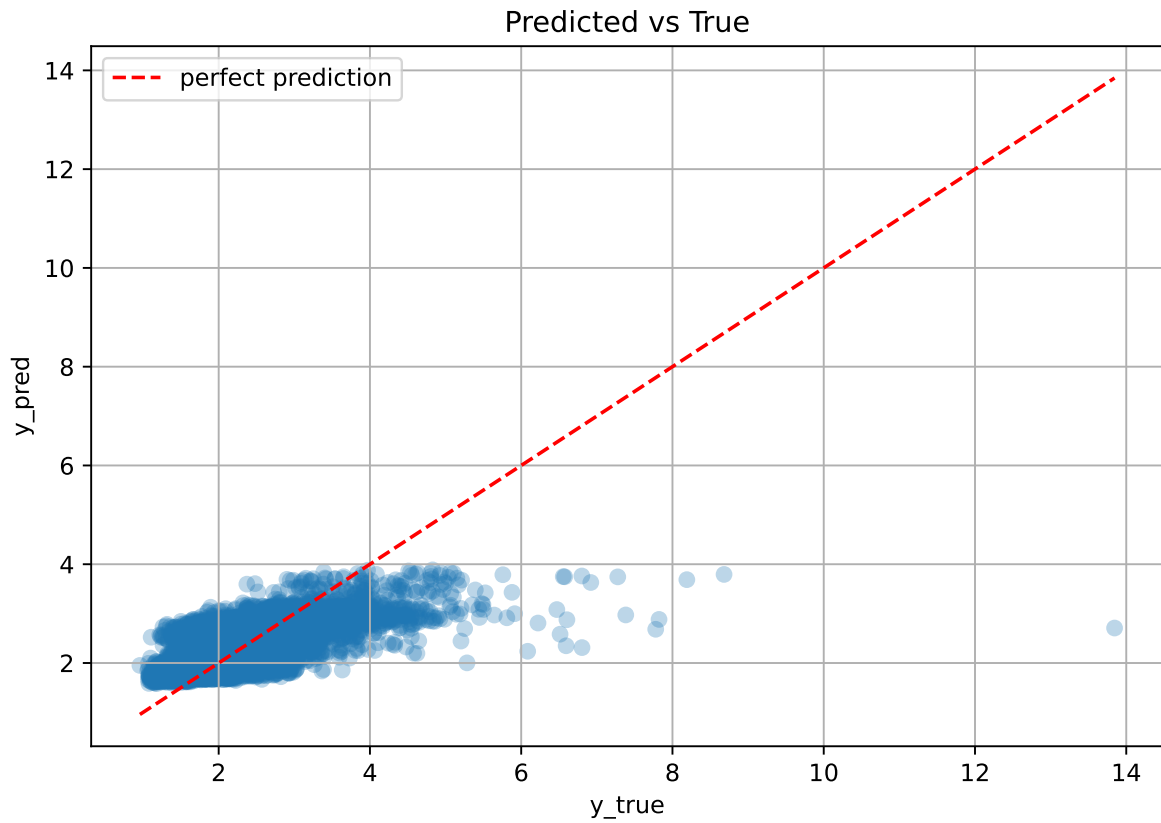
Rank 8: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.01734264217659695, 'reg__l2_leaf_reg': 10.64725218143674, 'reg__bagging_temperature': 1.4688699298424241, 'reg__random_strength': 3.680843650990199, 'reg__subsample': 0.8940364971626991, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8960845787931293, 'reg__border_count': 97, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5511.796138405804}

Rank 9: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.010885839292862949, 'reg__l2_leaf_reg': 28.94388572288884, 'reg__bagging_temperature': 0.9618350329356145, 'reg__random_strength': 4.419603538903862, 'reg__subsample': 0.846510987421108, 'reg__min_data_in_leaf': 32, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.8997984819389387, 'reg__border_count': 67, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 4787.024906330231}

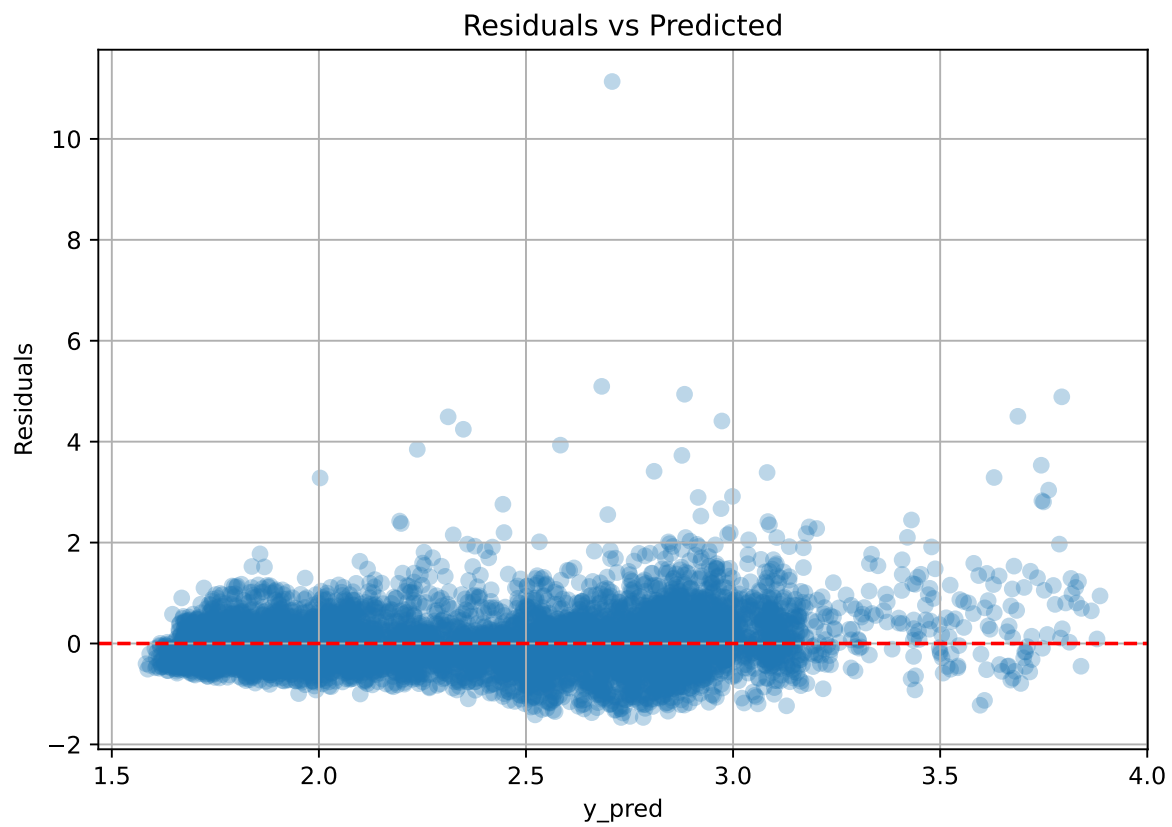
Rank 10: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.015858819323449975, 'reg__l2_leaf_reg': 9.303721300904133, 'reg__bagging_temperature': 2.0303538827353043,

```
'reg__random_strength': 4.991686662541303, 'reg__subsample': 0.8689687983075775,  
'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 8, 'reg__rsm':  
0.8825737137529961, 'reg__border_count': 93, 'reg__leaf_estimation_method': 'Newton',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_tempera-  
ture': 5722.494190990466}
```

3.6.2 Scatter



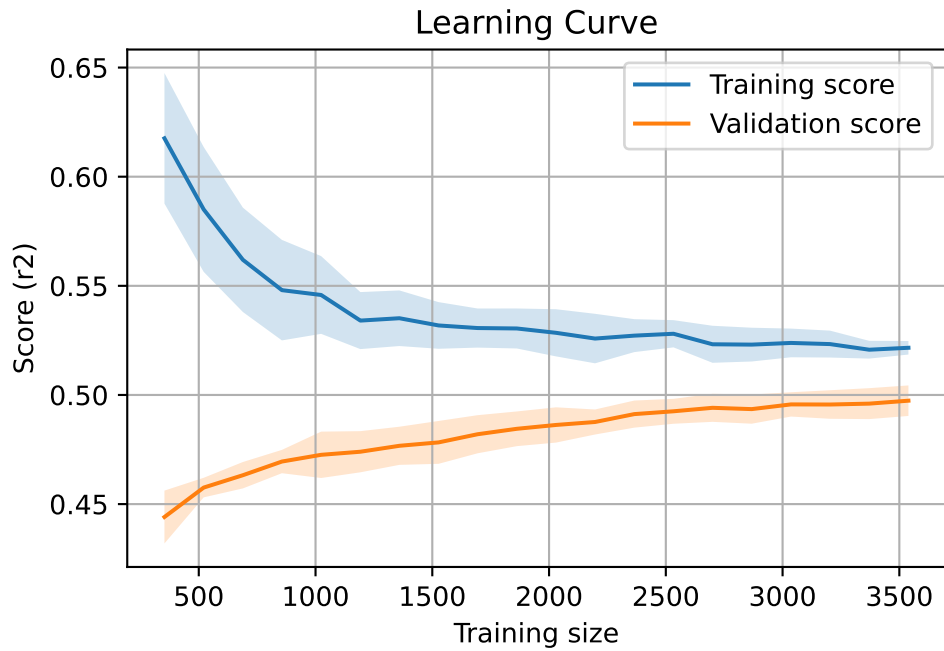
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7224, `Len(training)` : 5057,

`Len(training_real)` : 3540, `Len(test_of_training)` : 1517, `Len(test)` : 2167



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.422	0.414	0.013	0.374	0.375	-0.038	0.908	1.022	nan
LGBM	0.49	0.468	0.012	0.416	0.439	-0.03	0.937	1.046	0.4684
CatBoost	0.523	0.498	0.008	0.446	0.451	-0.047	0.905	1.048	0.4985

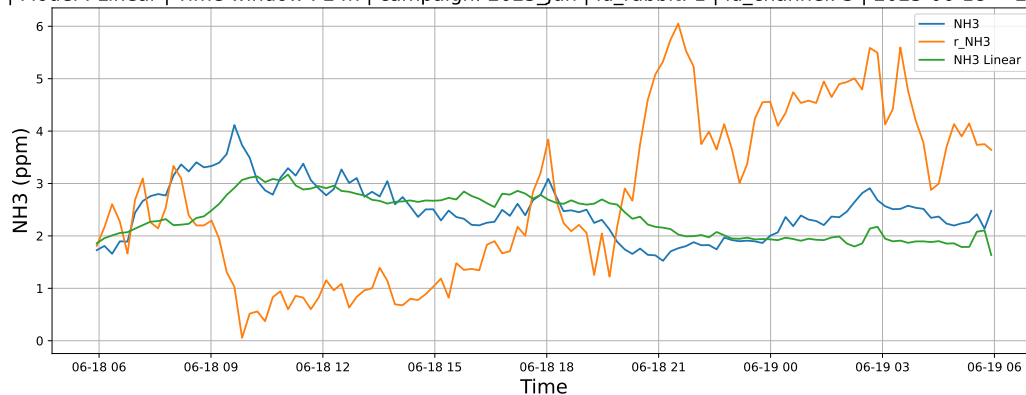
4 Plot

4.1 Standalone plots

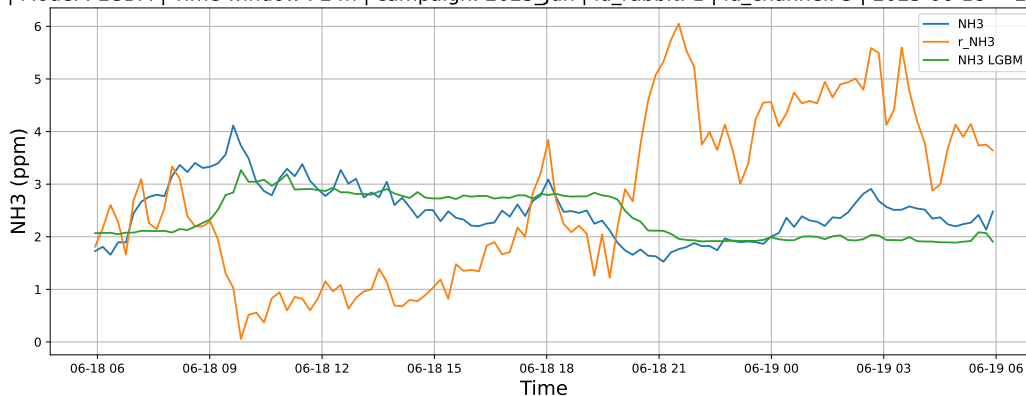
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

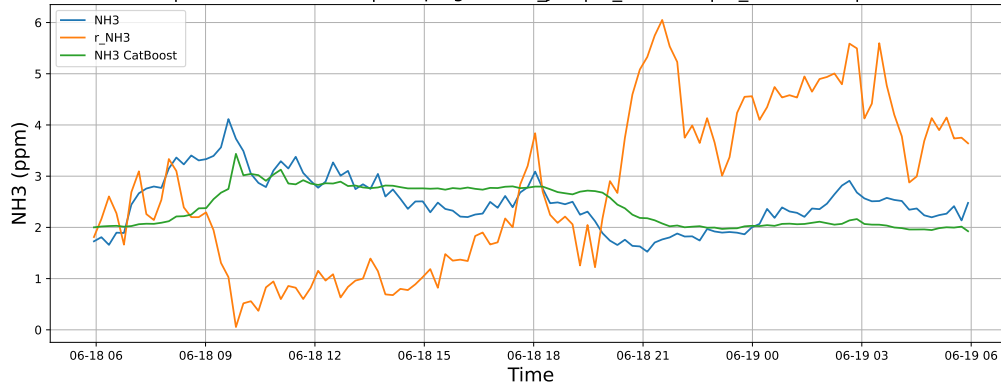
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

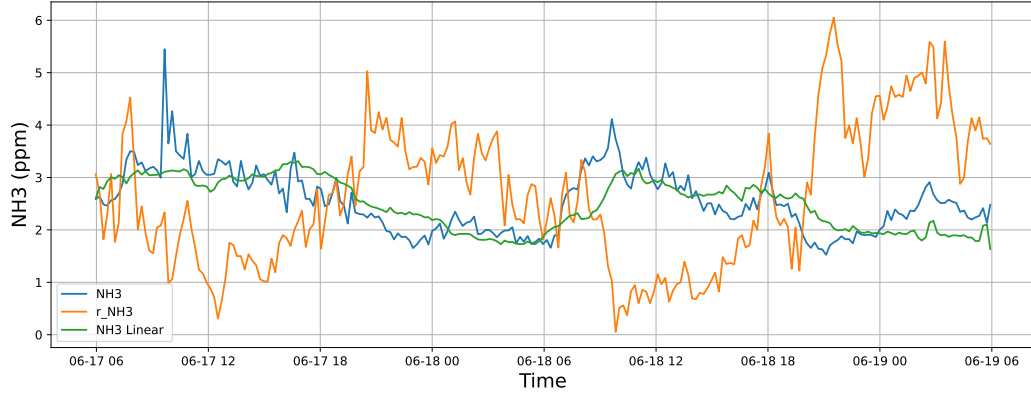


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

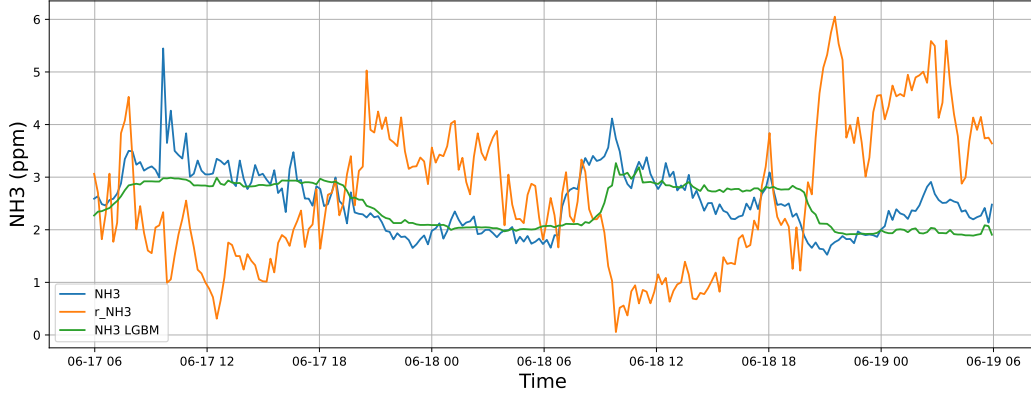


4.1.1.2 Time window : 48

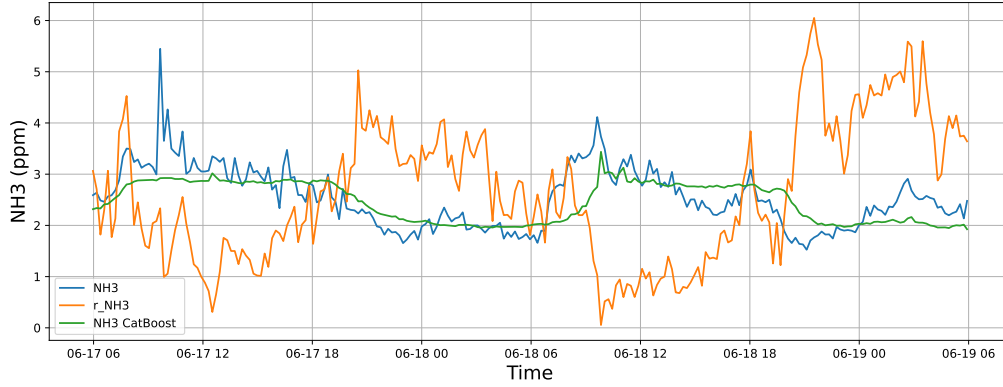
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

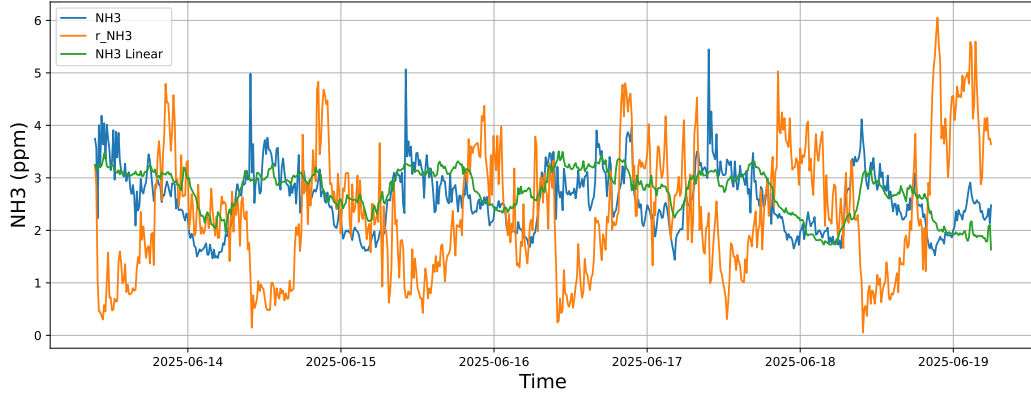


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

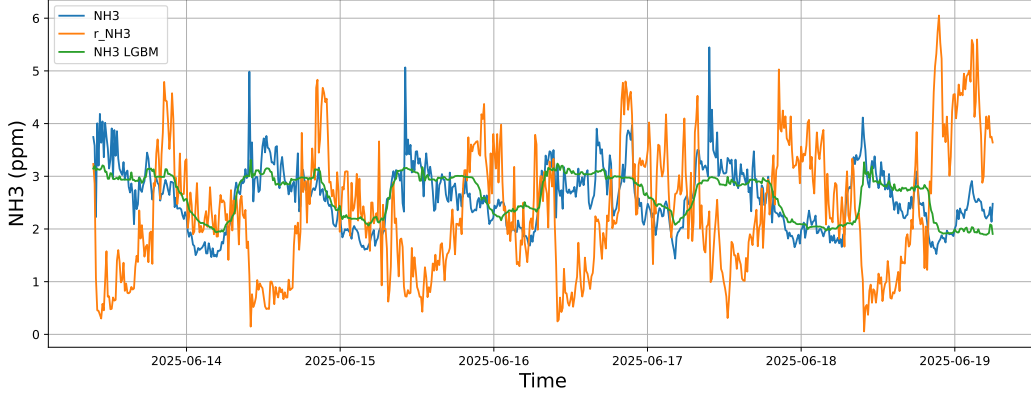


4.1.1.3 Time window : ALL

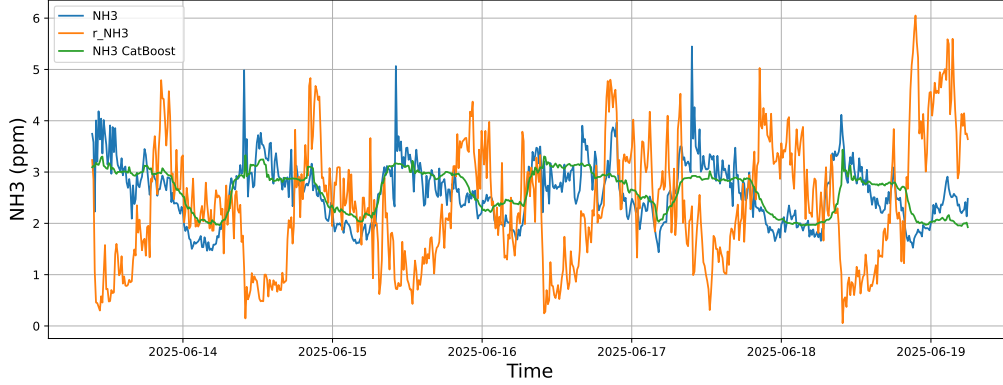
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



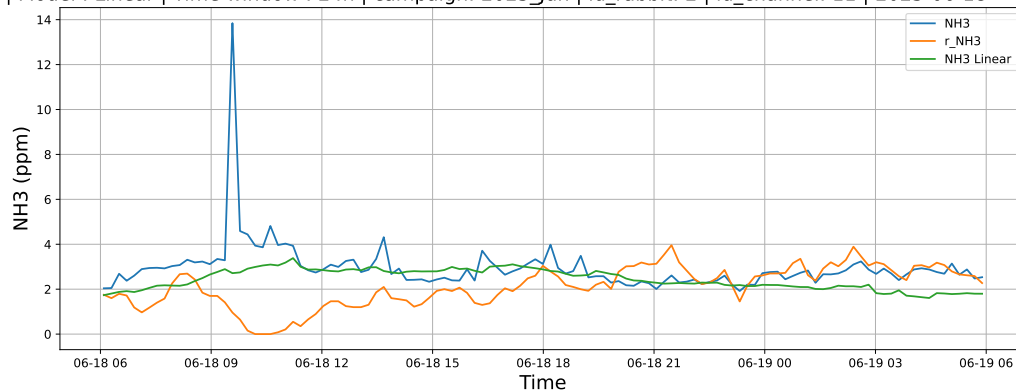
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



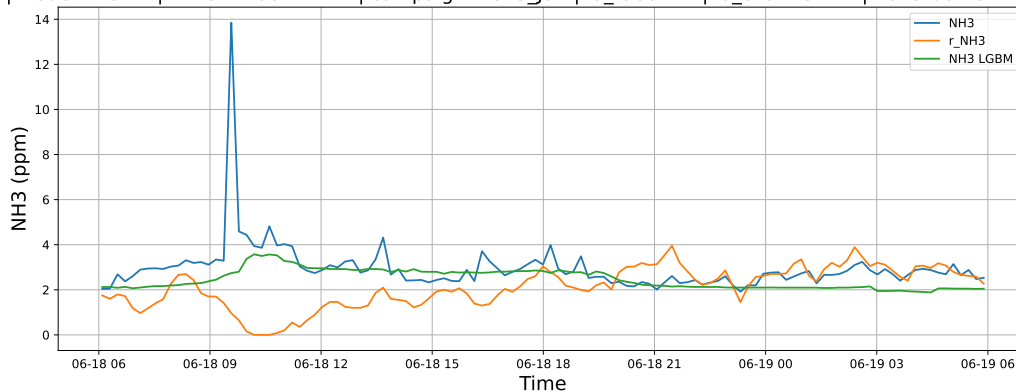
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

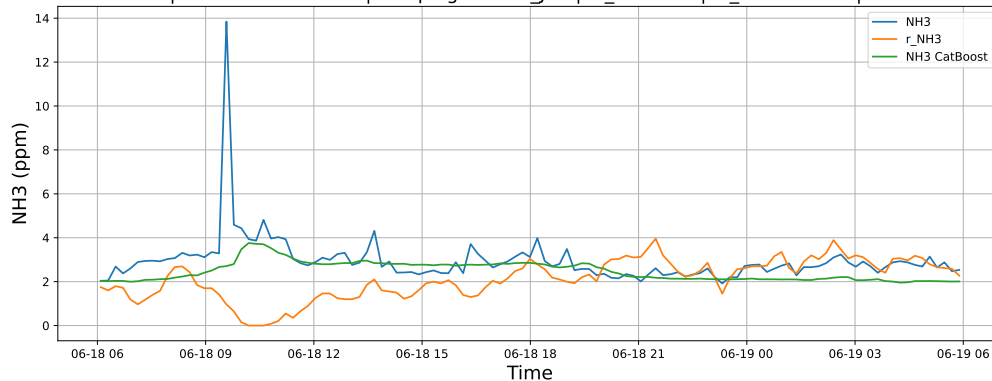
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

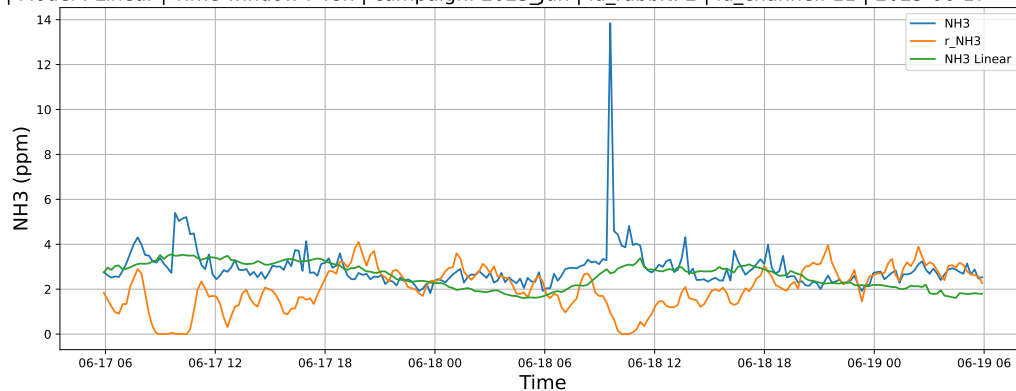


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

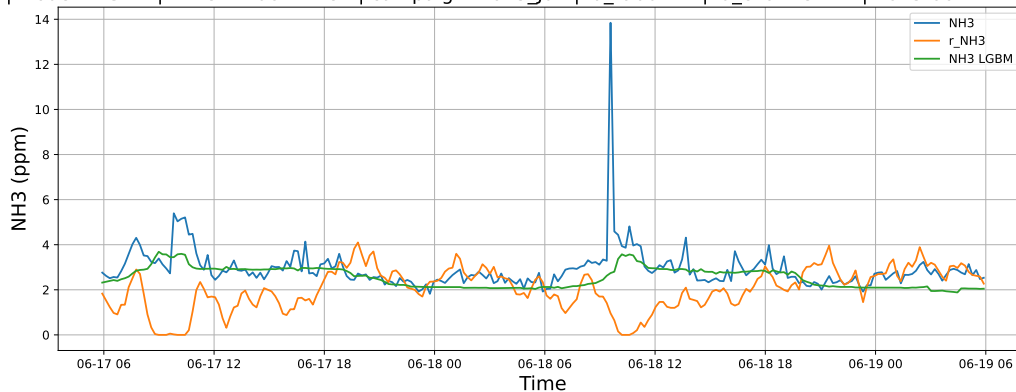


4.1.2.2 Time window : 48

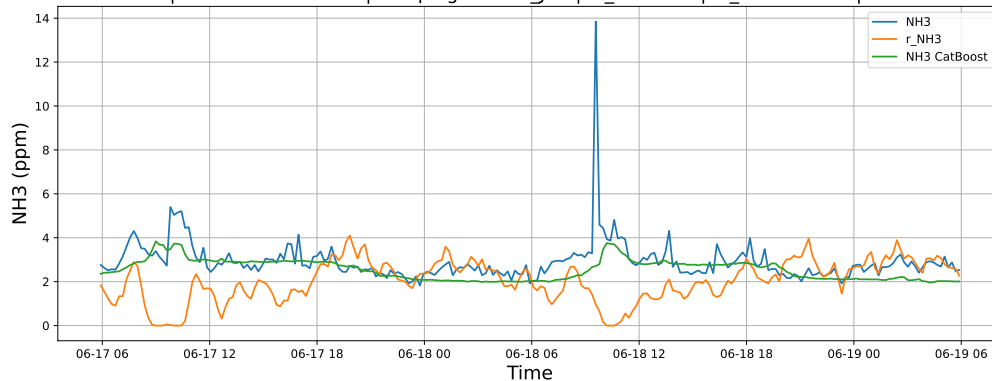
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

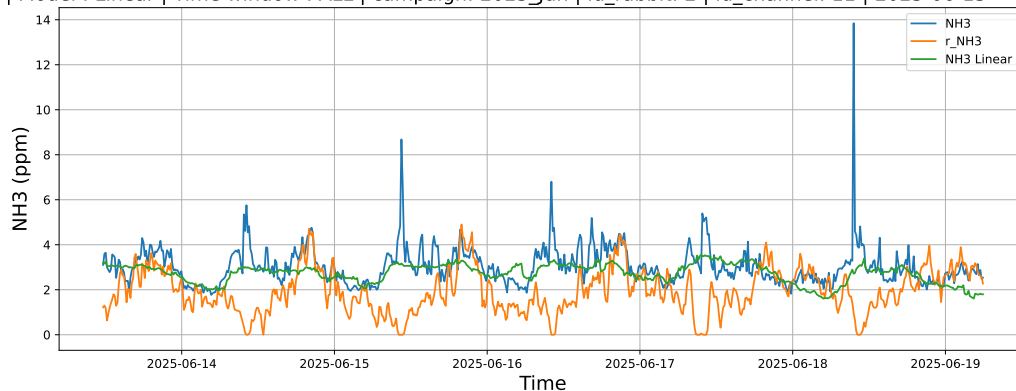


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

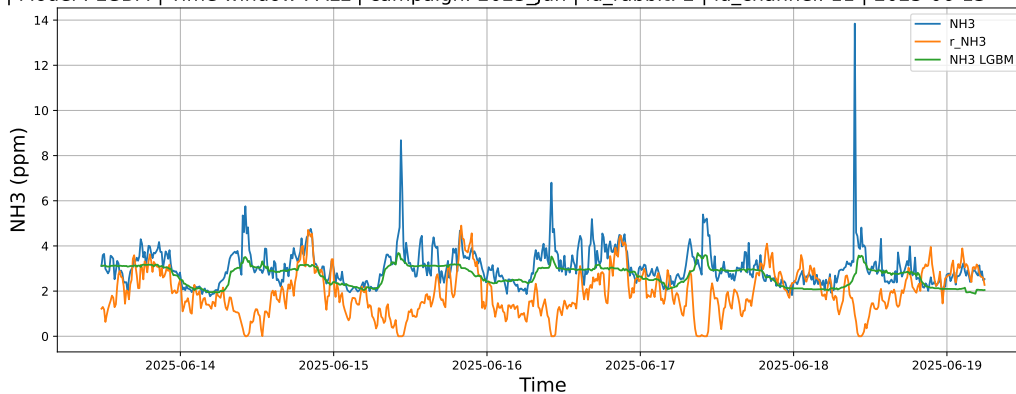


4.1.2.3 Time window : ALL

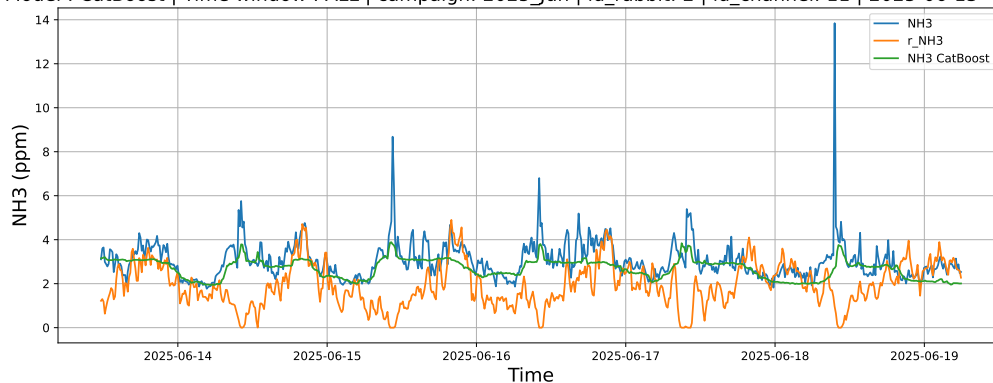
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



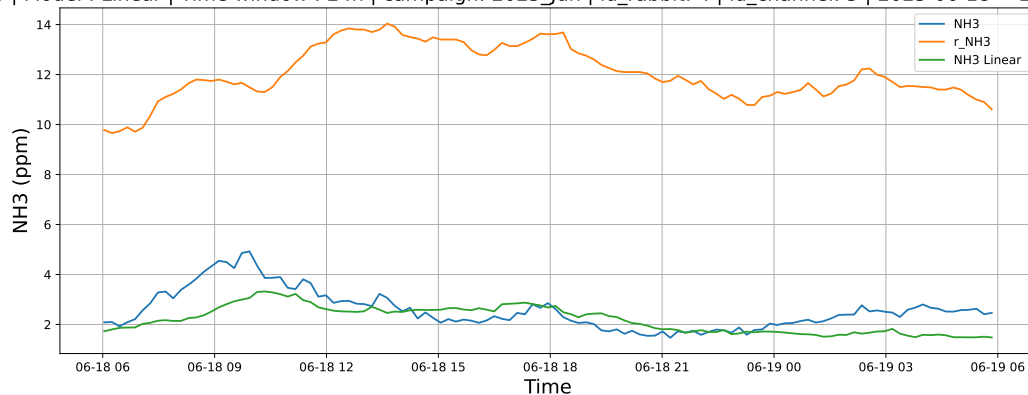
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



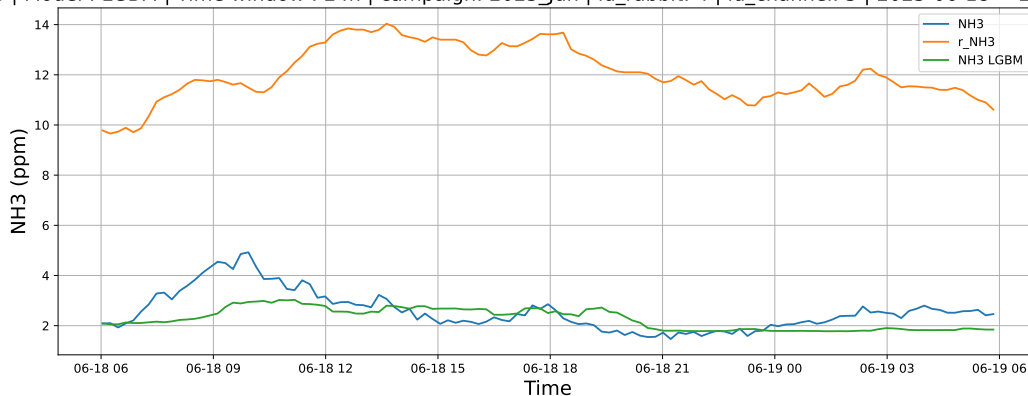
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

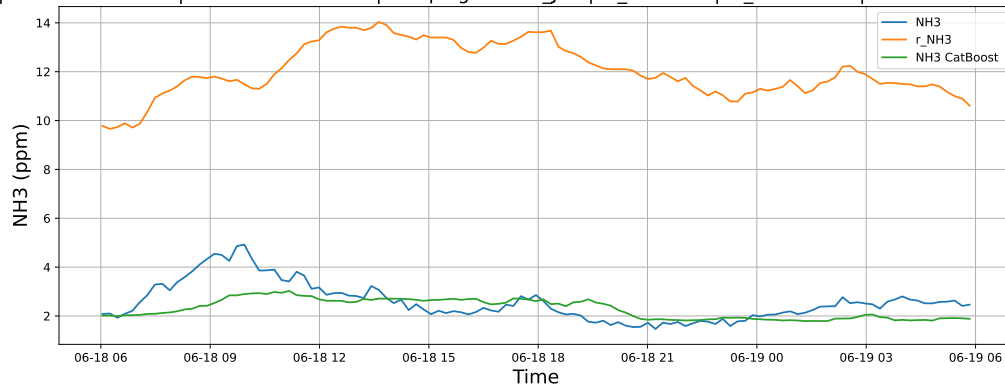
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

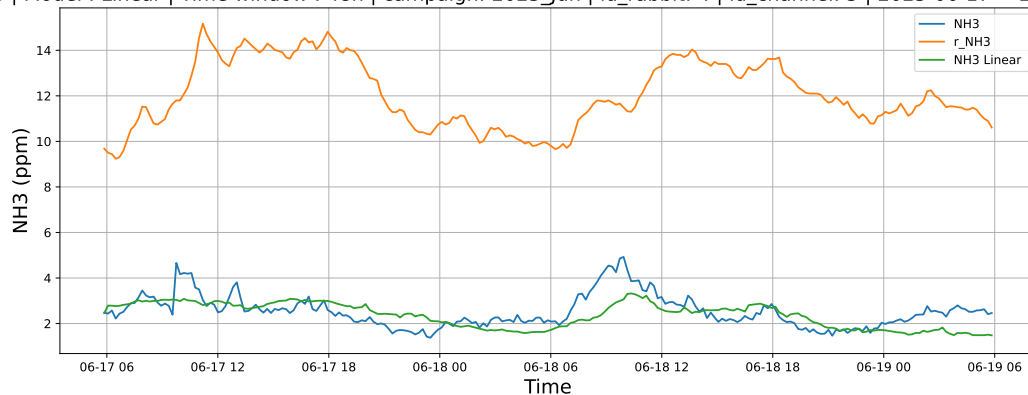


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

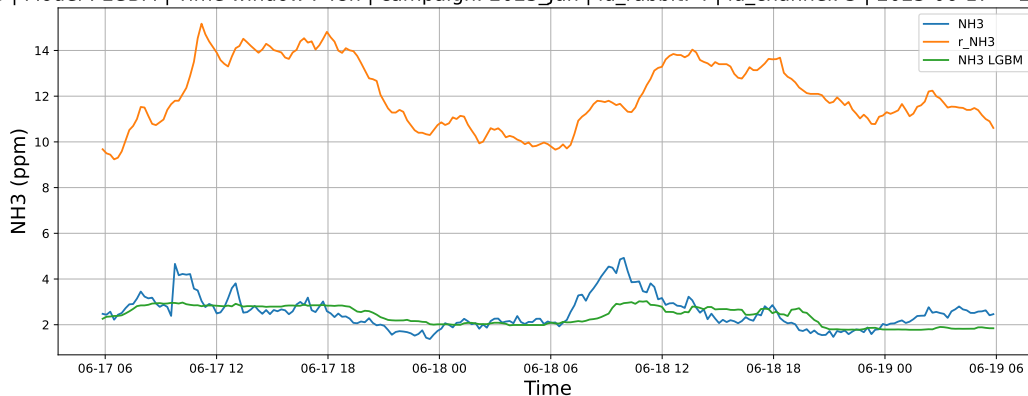


4.1.3.2 Time window : 48

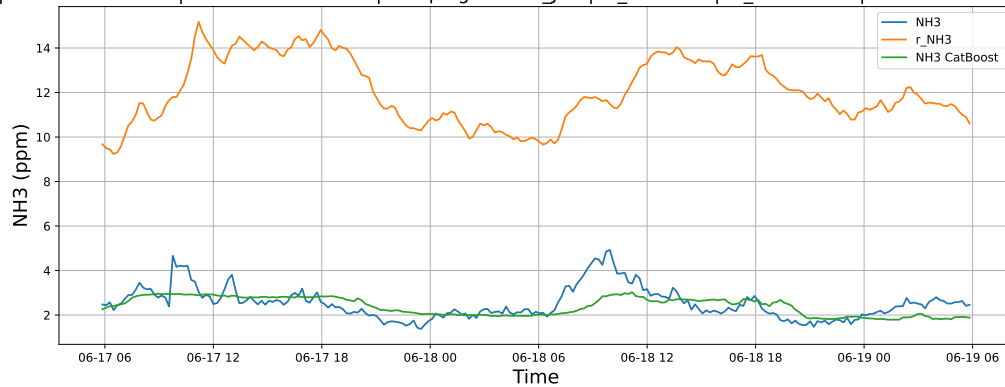
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

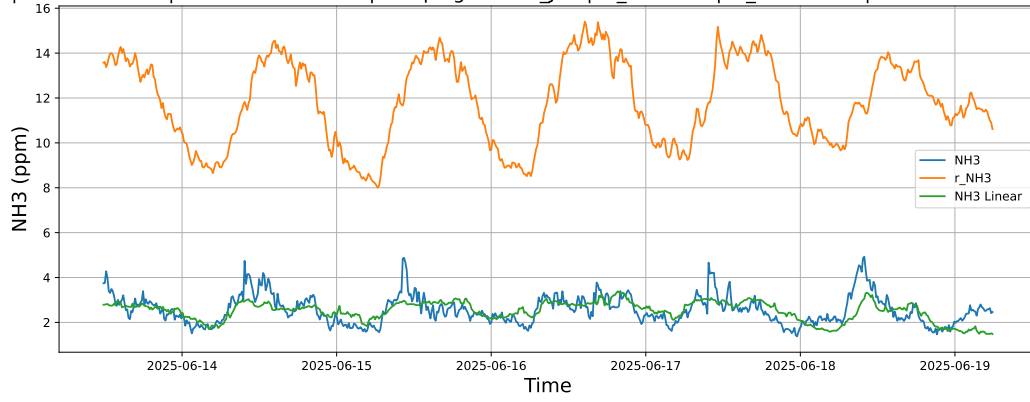


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

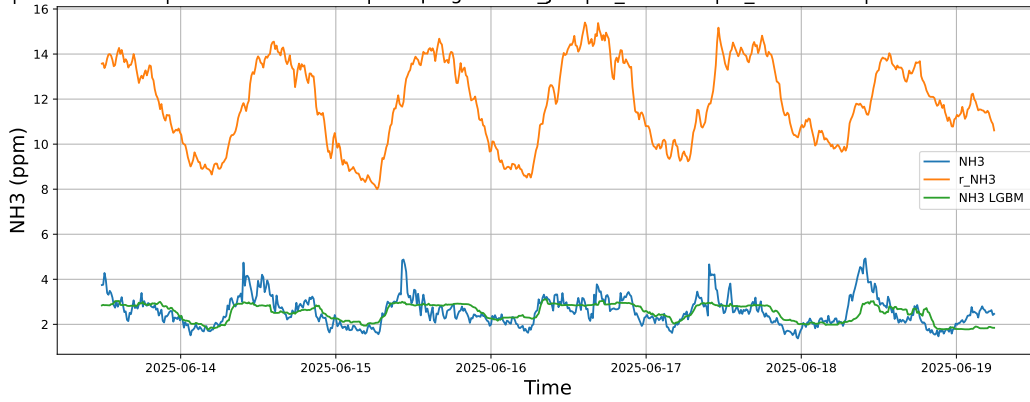


4.1.3.3 Time window : ALL

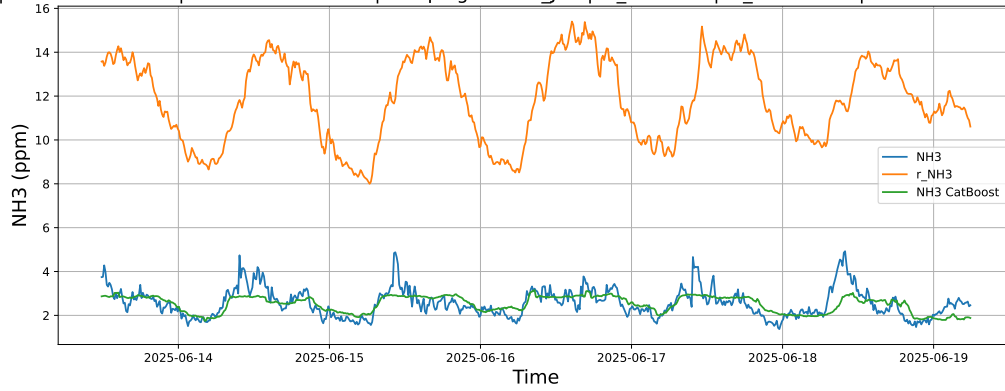
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



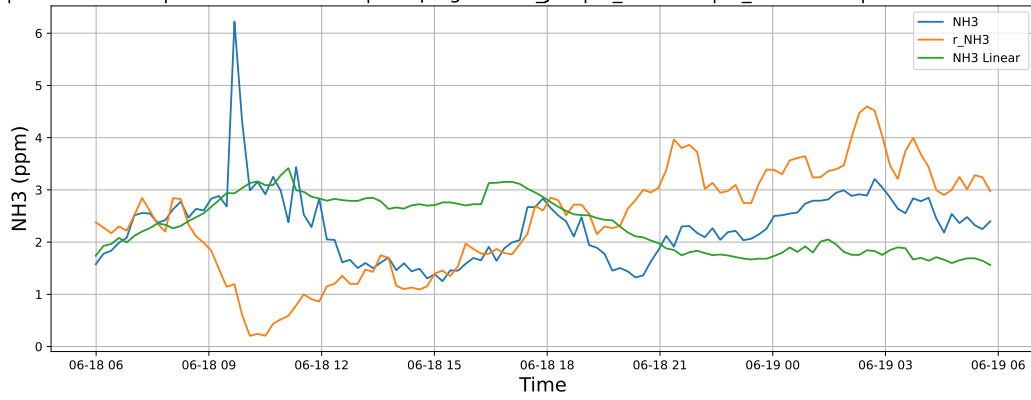
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



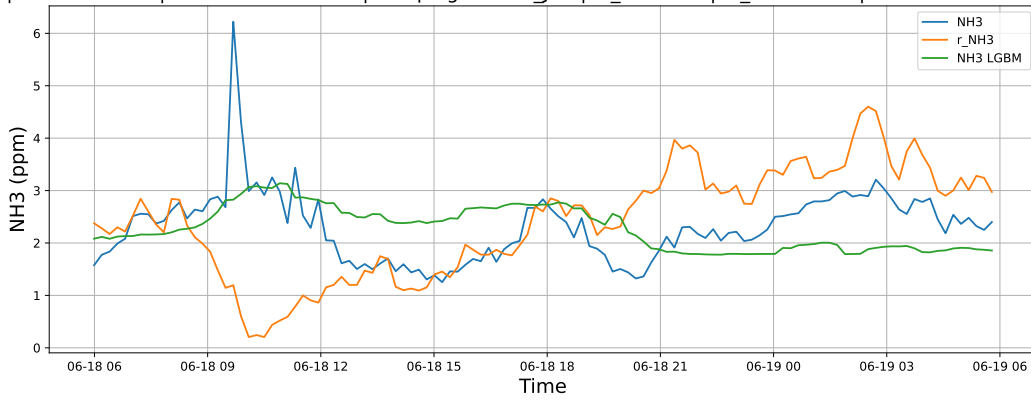
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

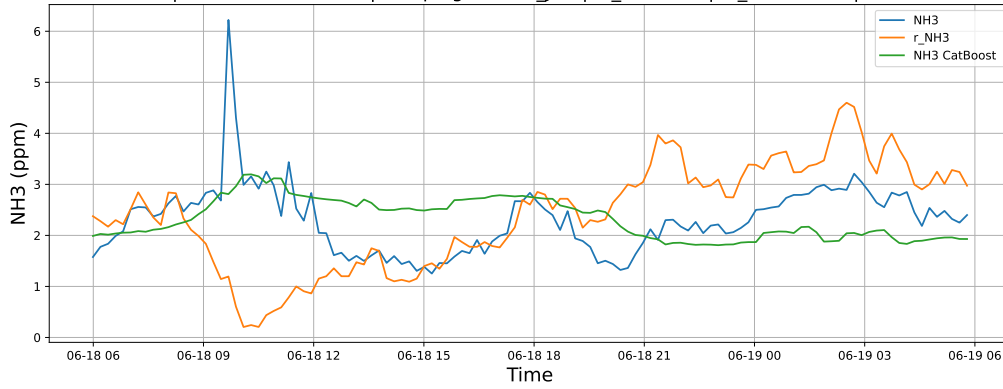
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

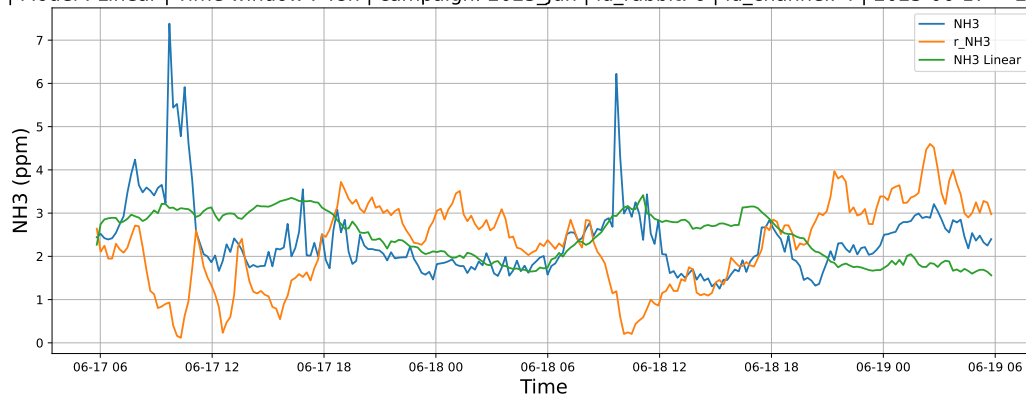


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

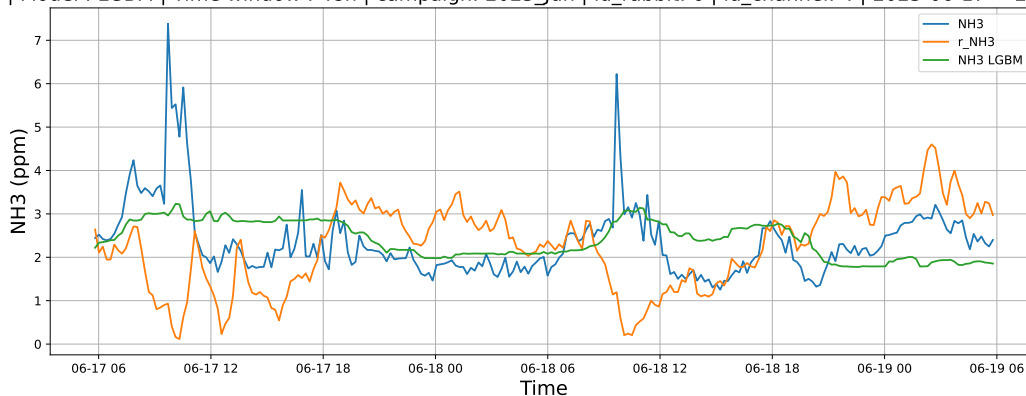


4.1.4.2 Time window : 48

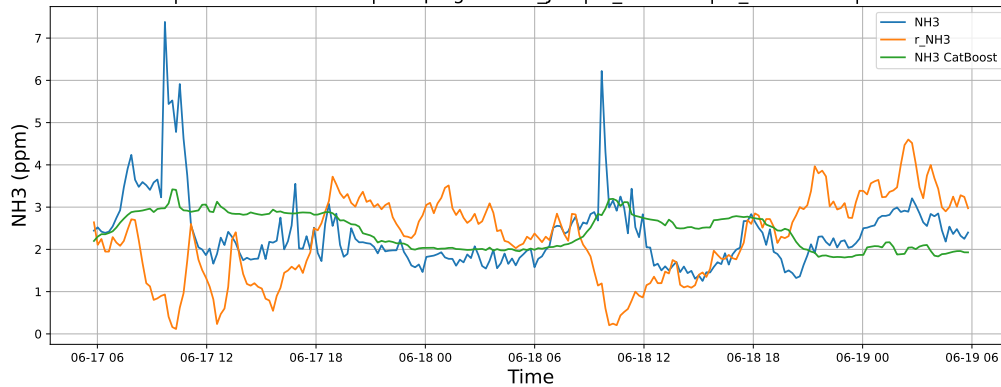
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

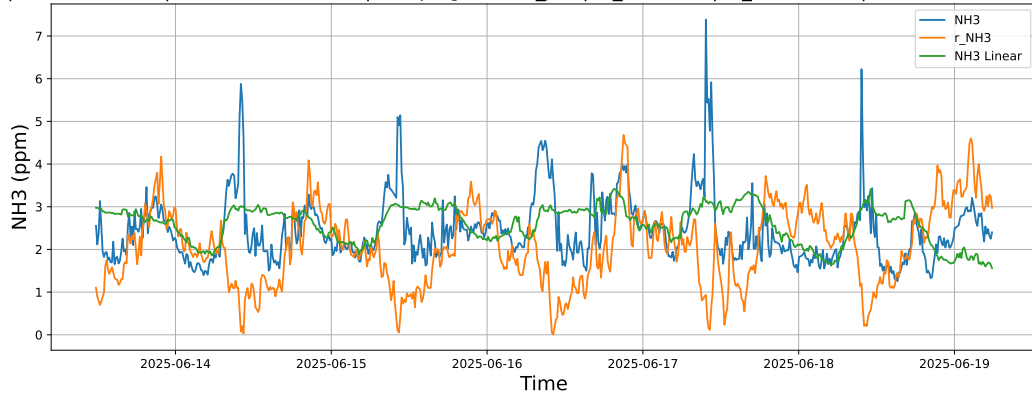


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

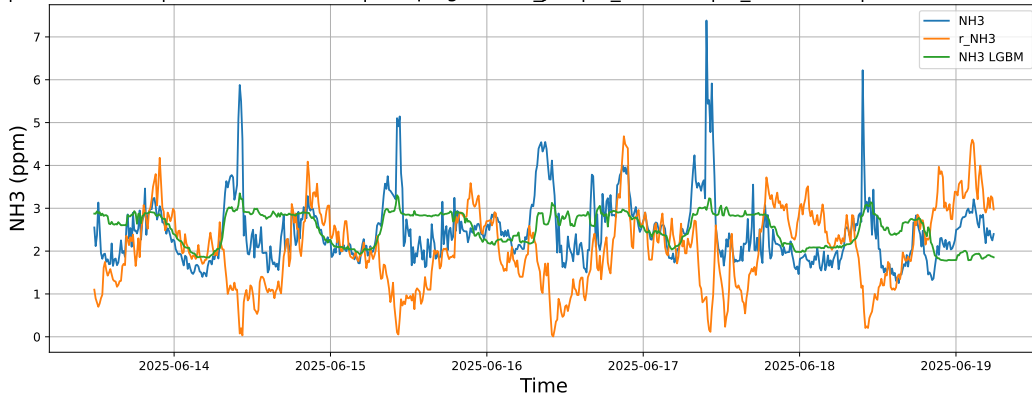


4.1.4.3 Time window : ALL

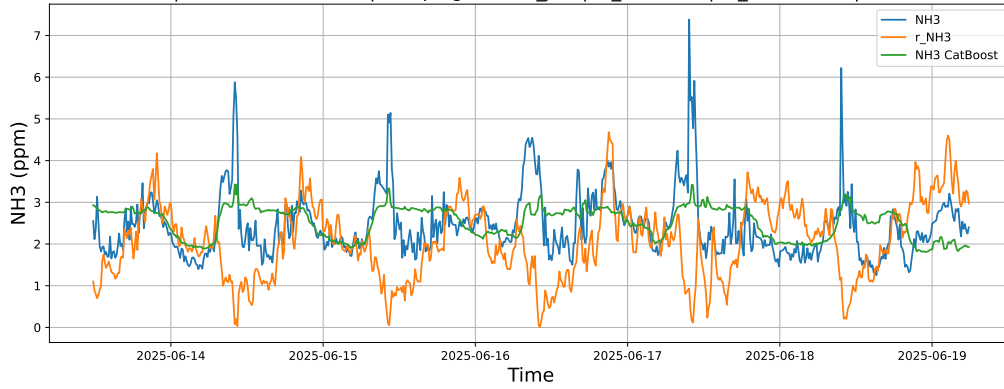
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



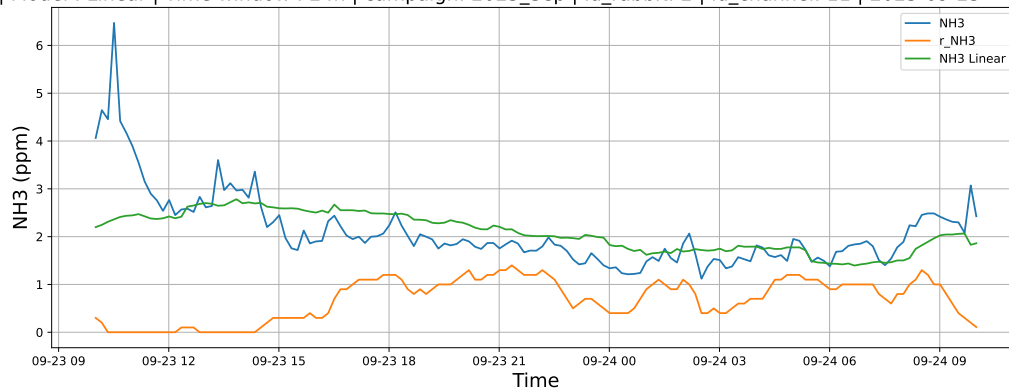
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



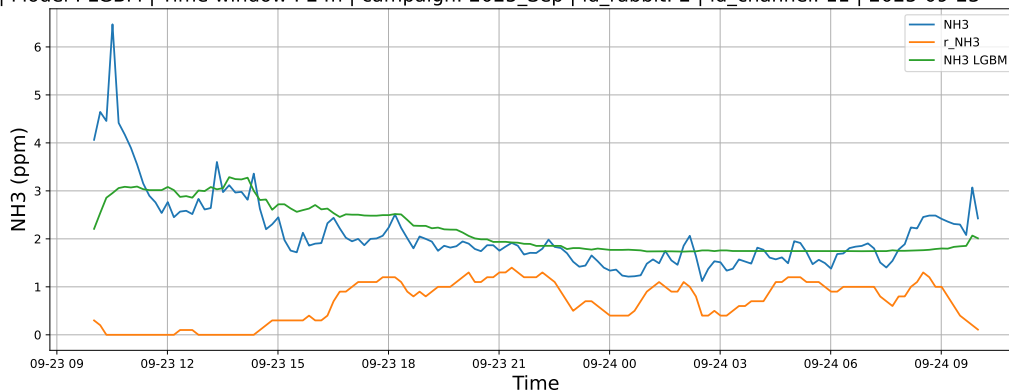
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

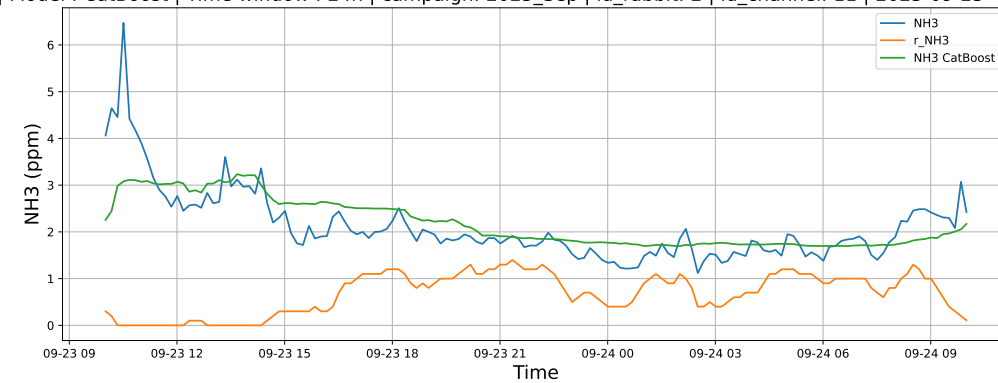
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

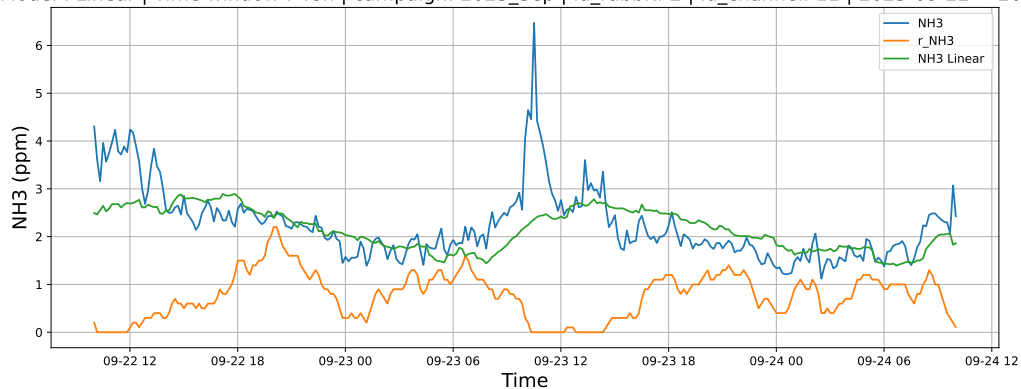


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

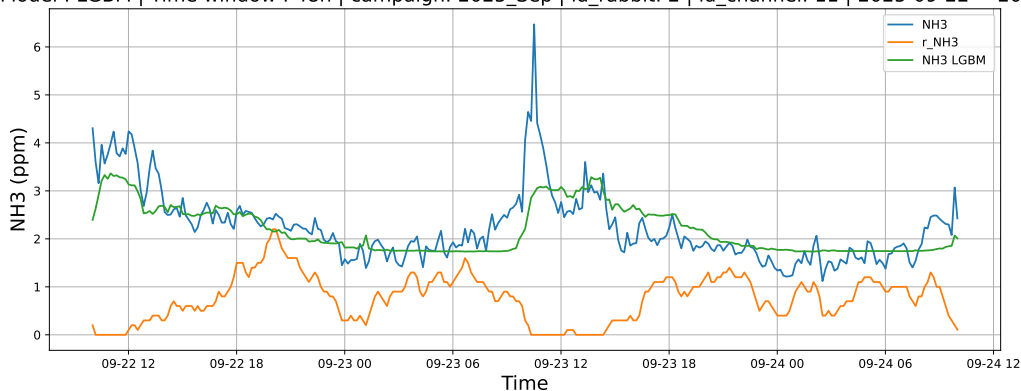


4.1.5.2 Time window : 48

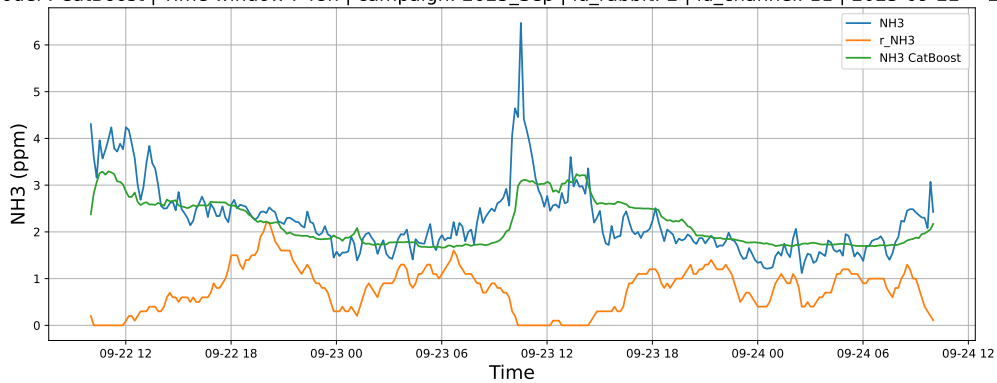
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

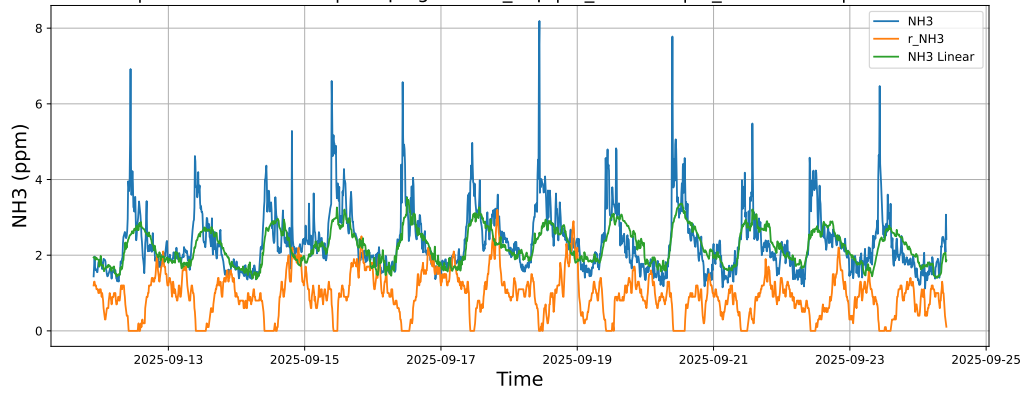


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

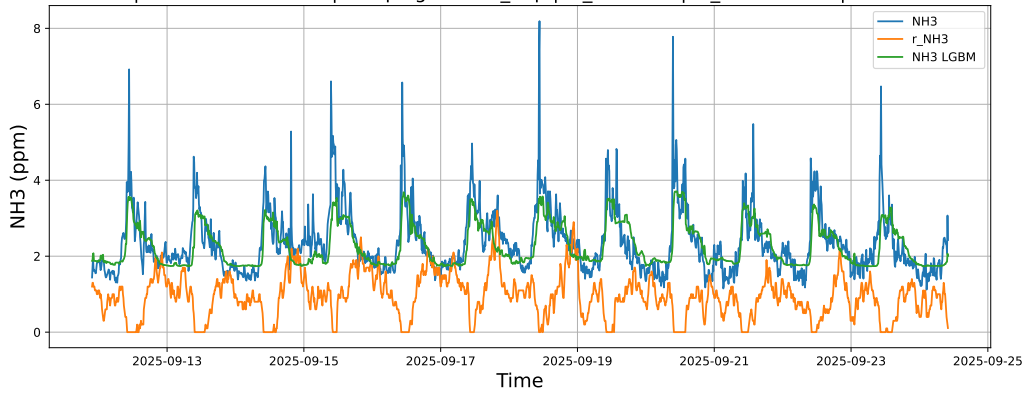


4.1.5.3 Time window : ALL

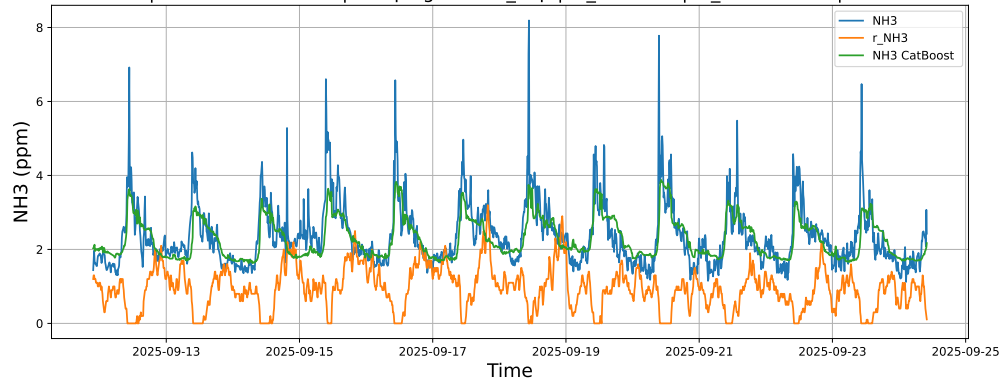
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



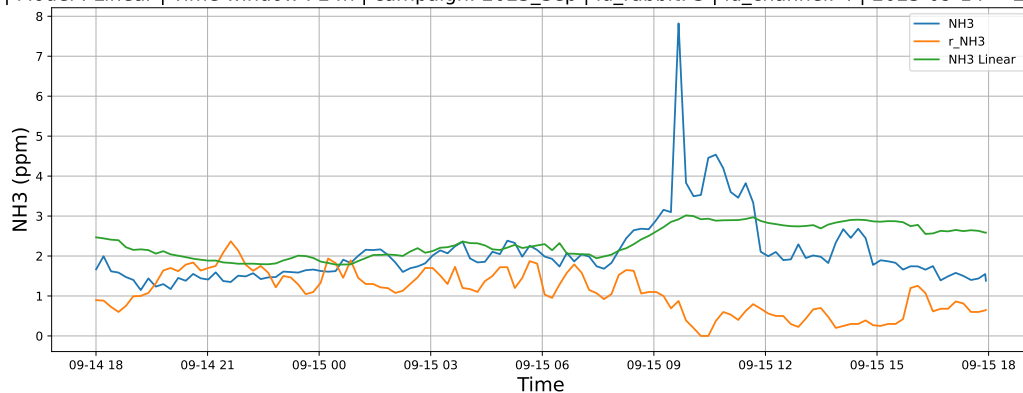
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



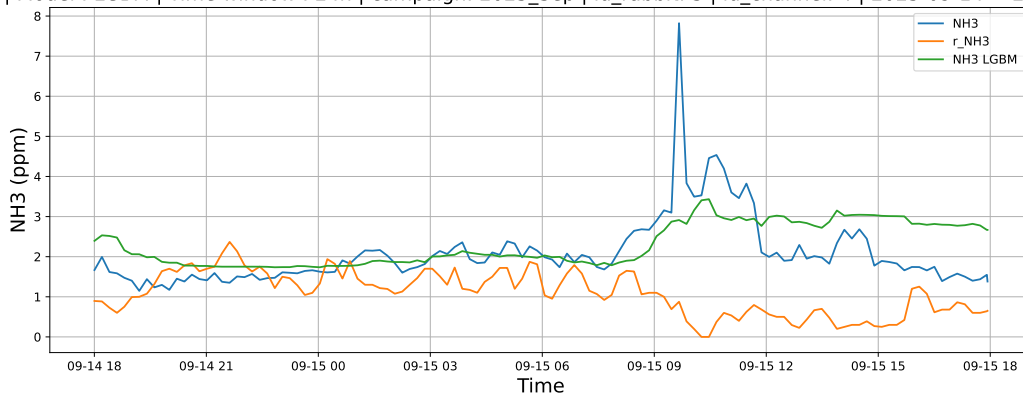
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

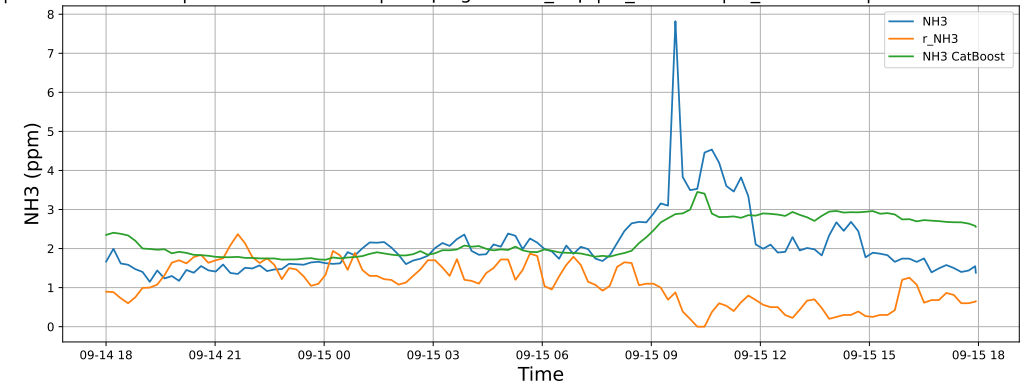
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

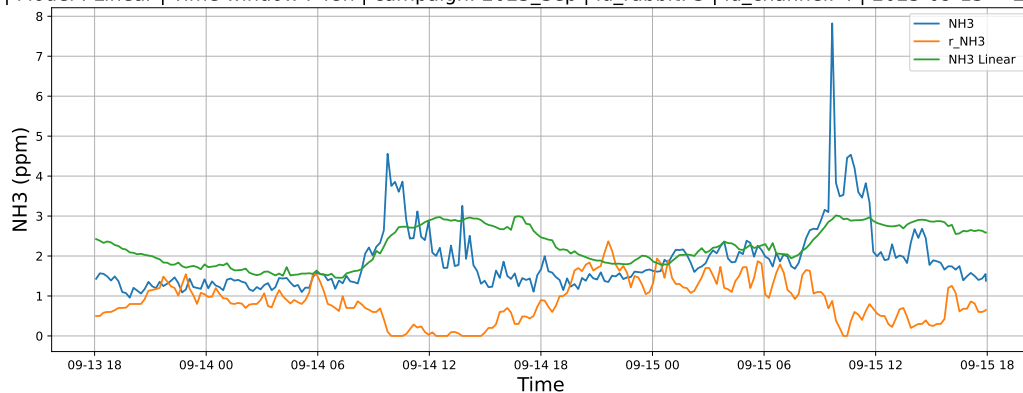


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

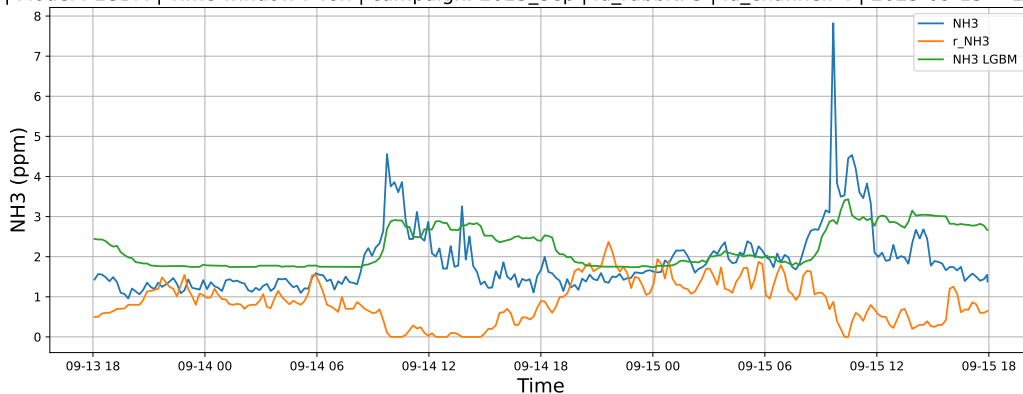


4.1.6.2 Time window : 48

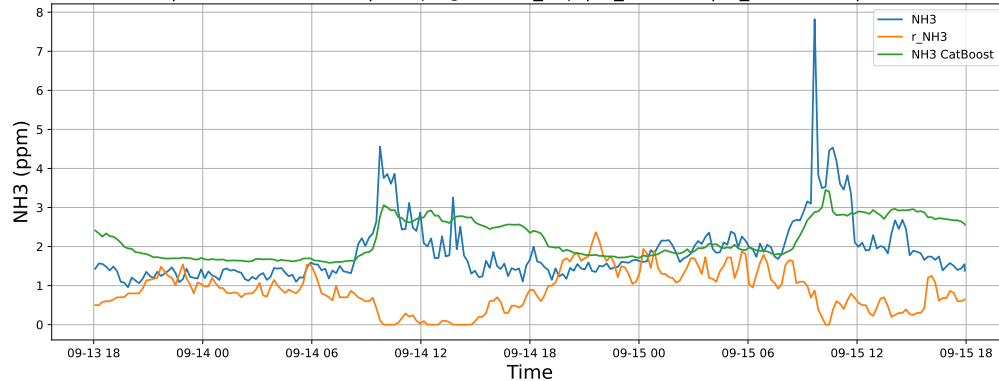
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

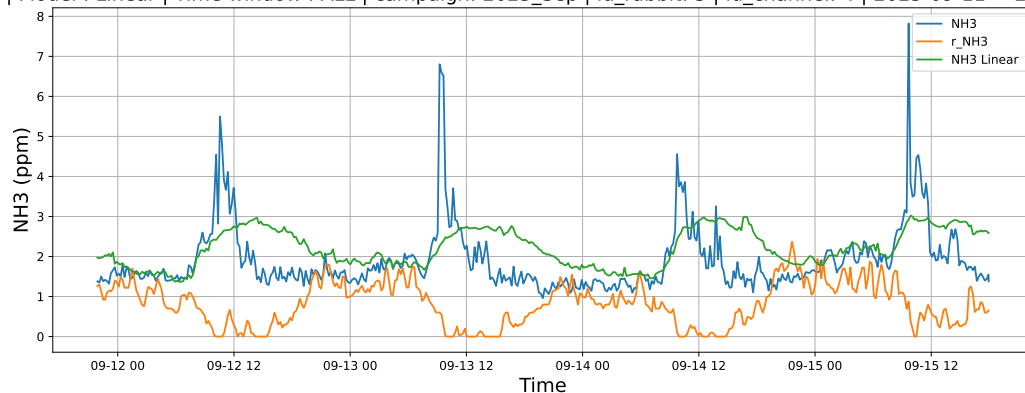


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

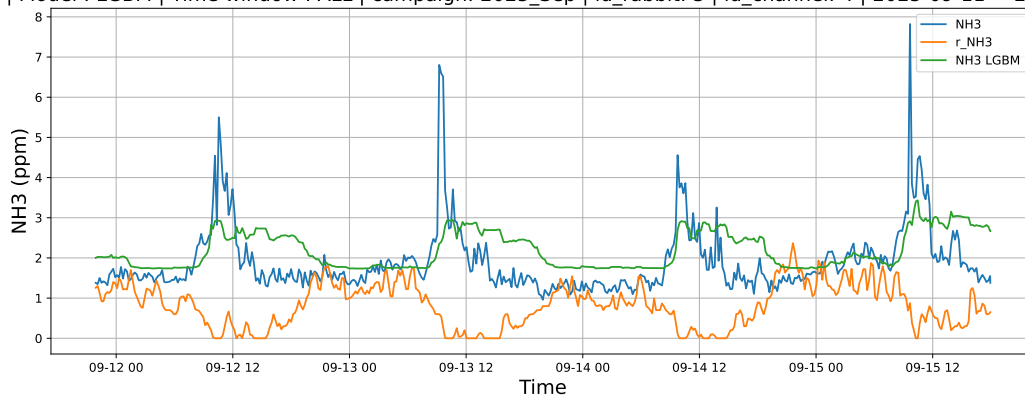


4.1.6.3 Time window : ALL

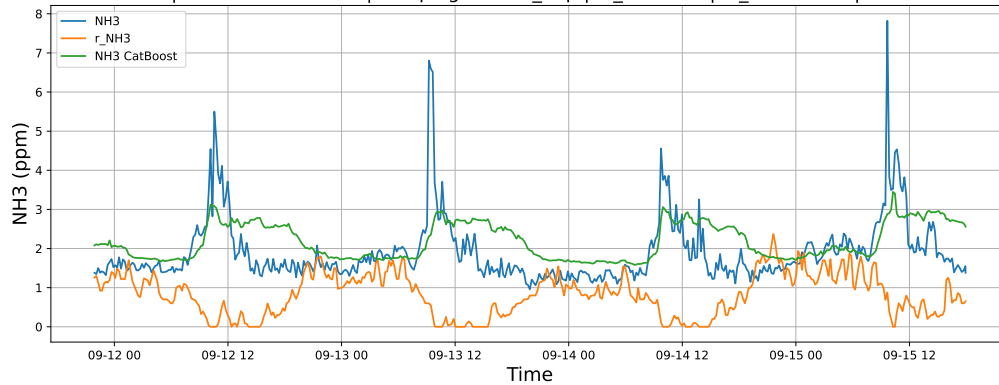
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



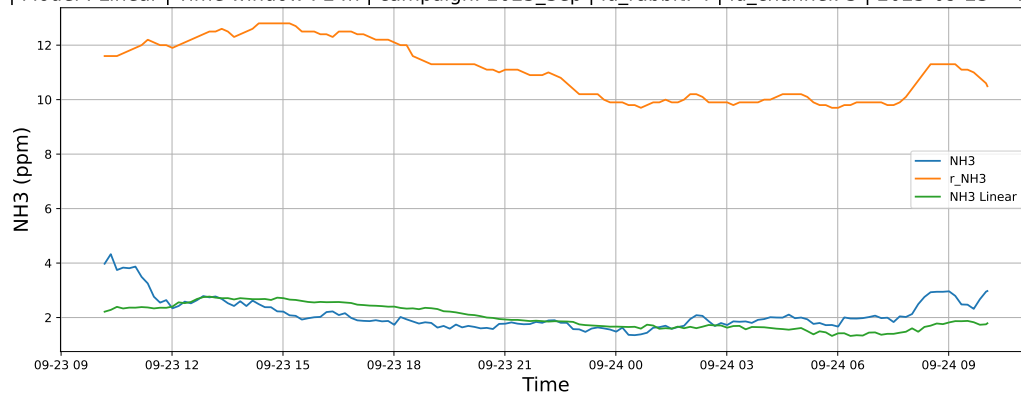
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



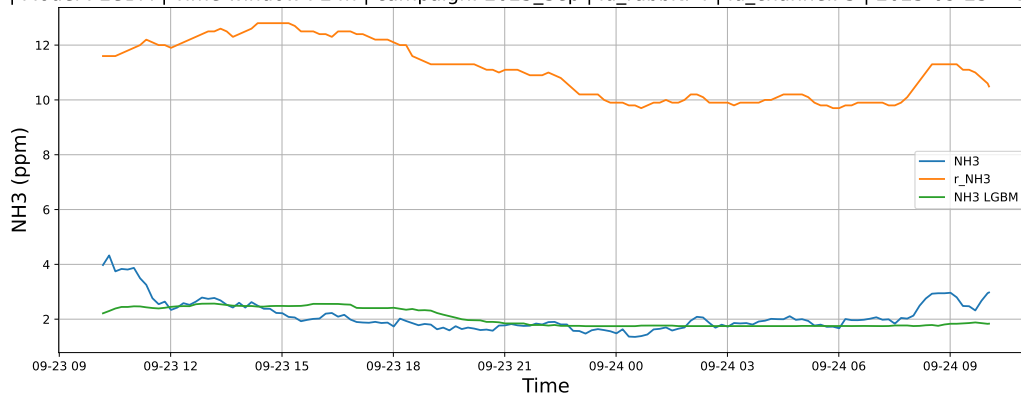
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

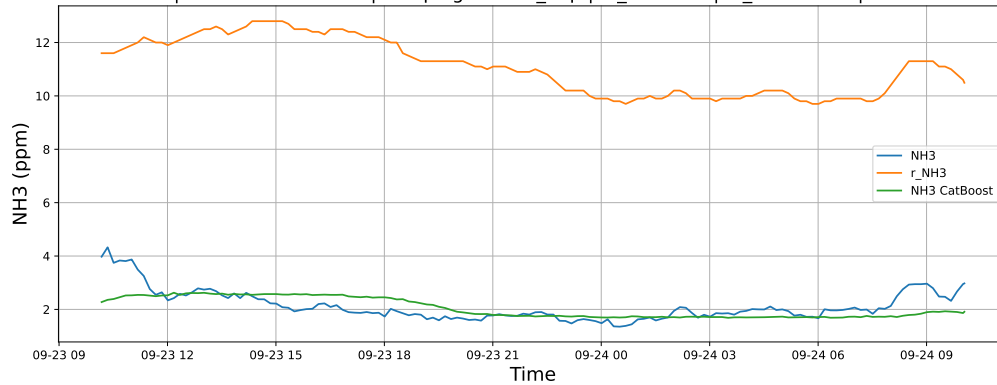
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

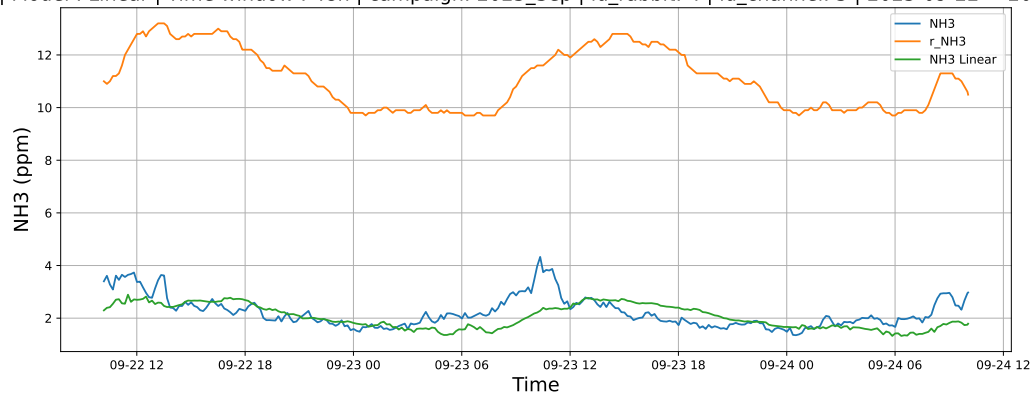


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

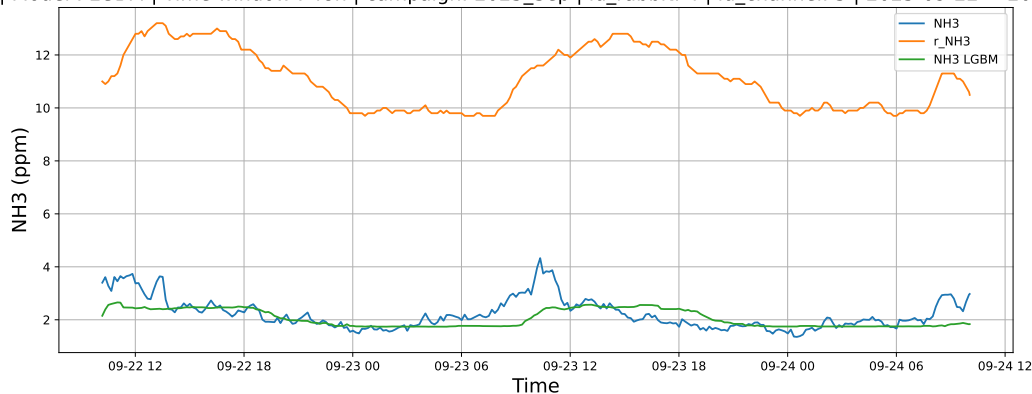


4.1.7.2 Time window : 48

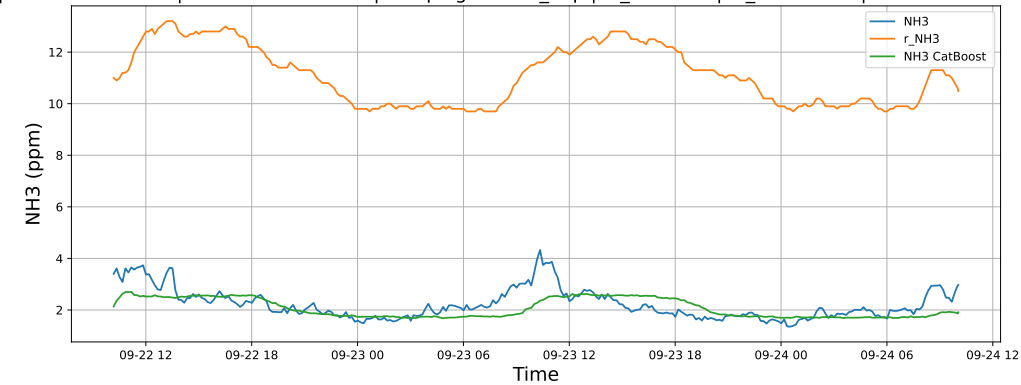
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

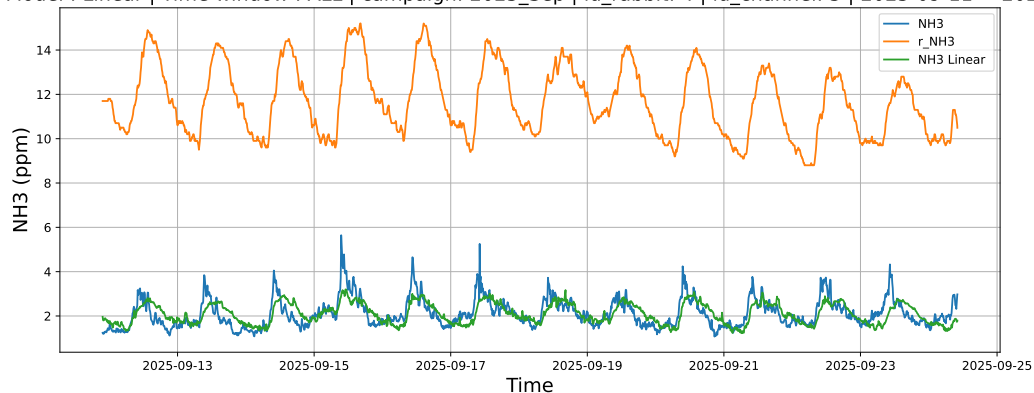


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

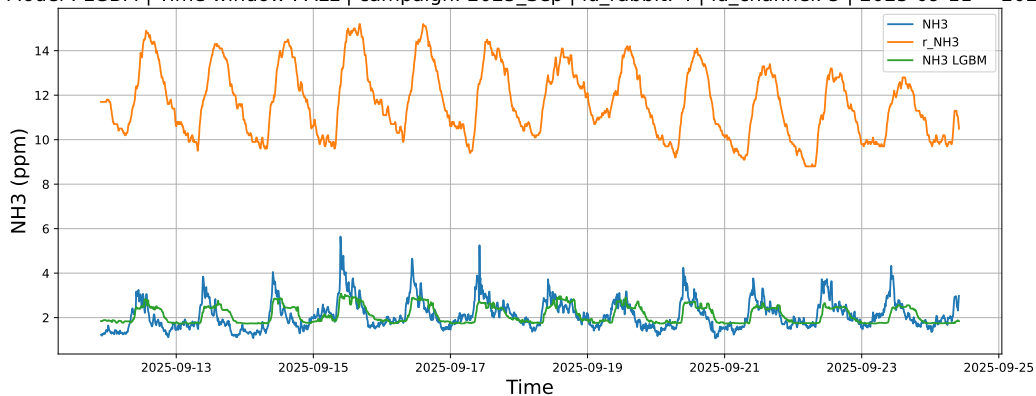


4.1.7.3 Time window : ALL

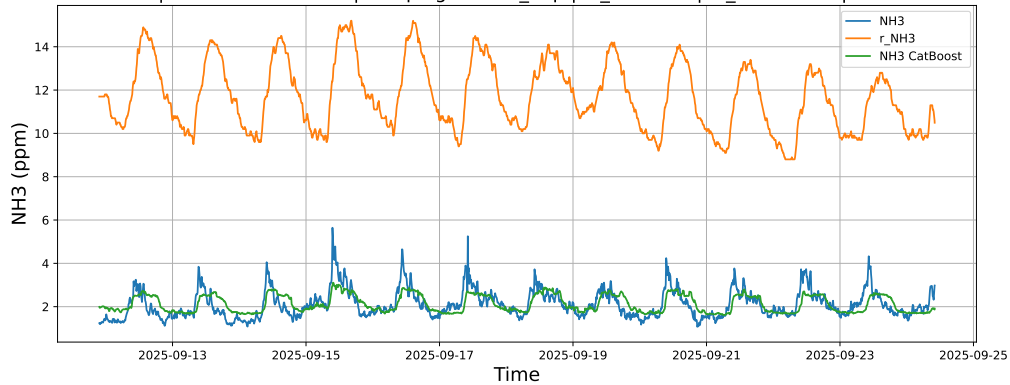
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



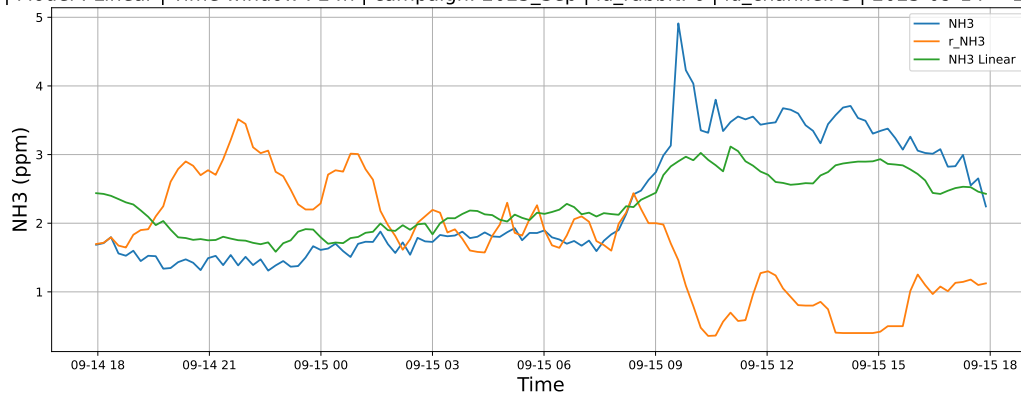
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



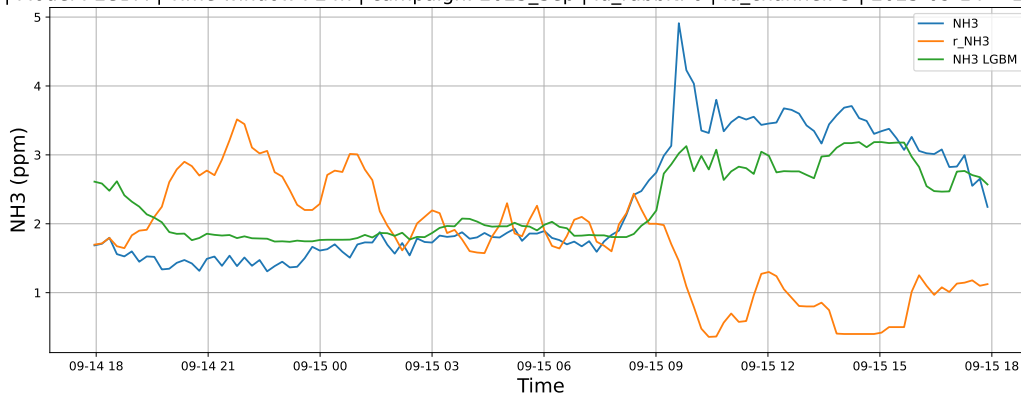
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

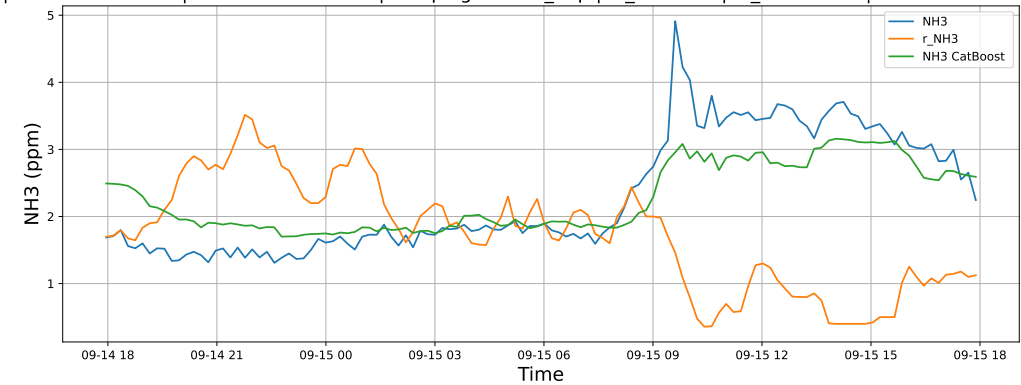
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

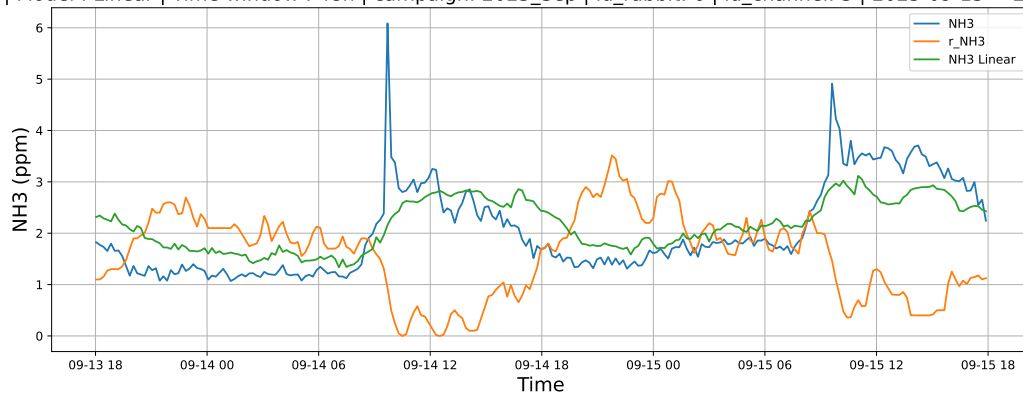


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

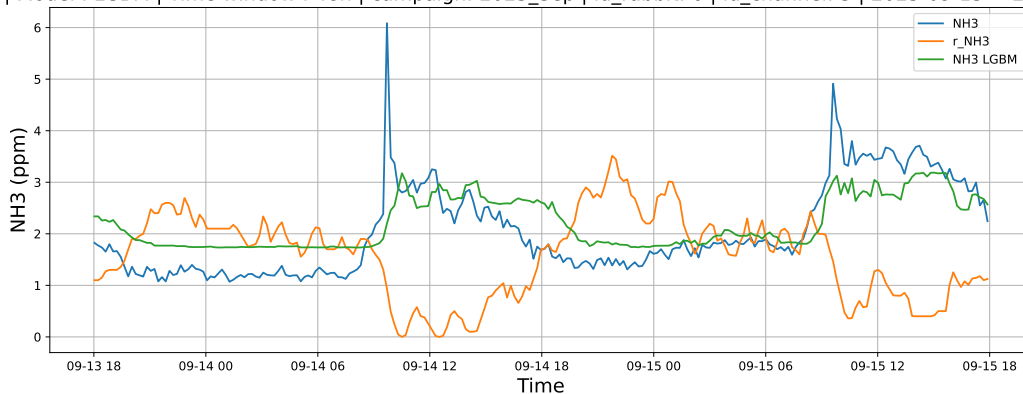


4.1.8.2 Time window : 48

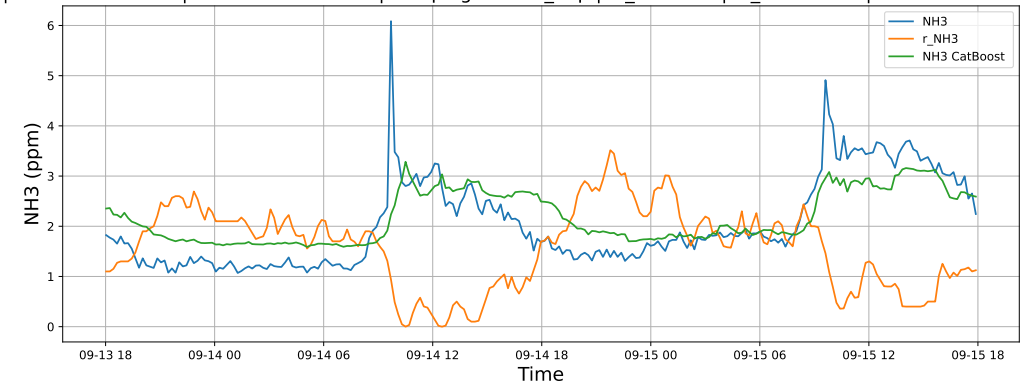
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

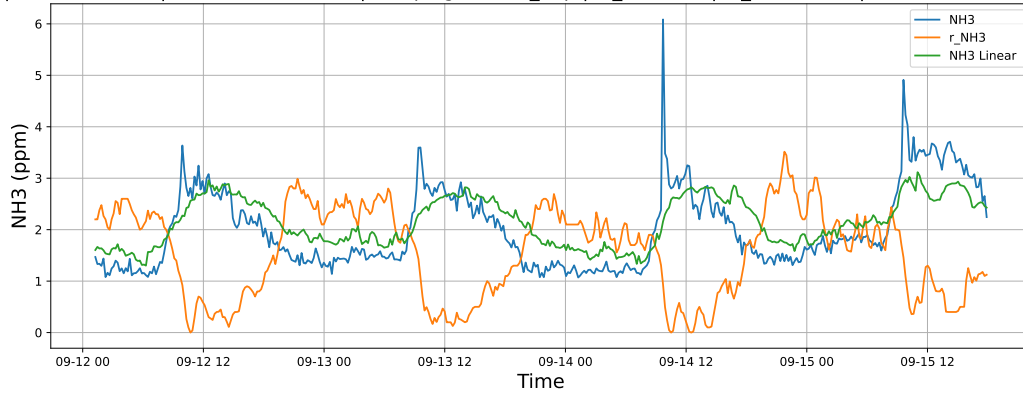


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

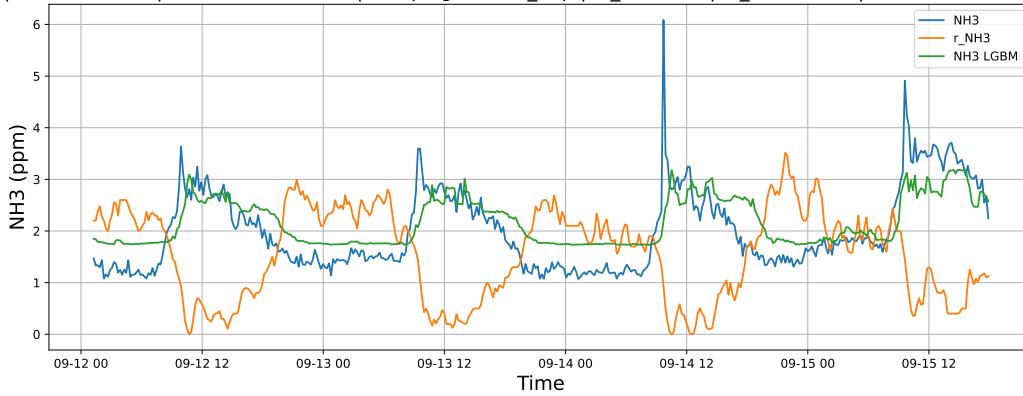


4.1.8.3 Time window : ALL

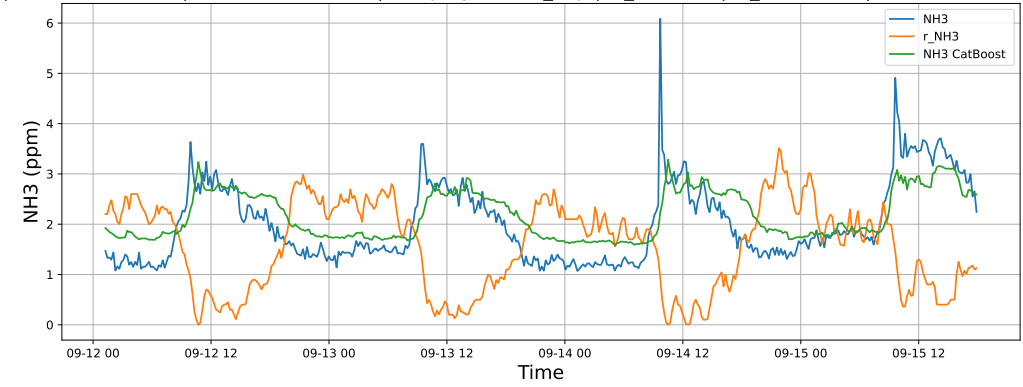
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15

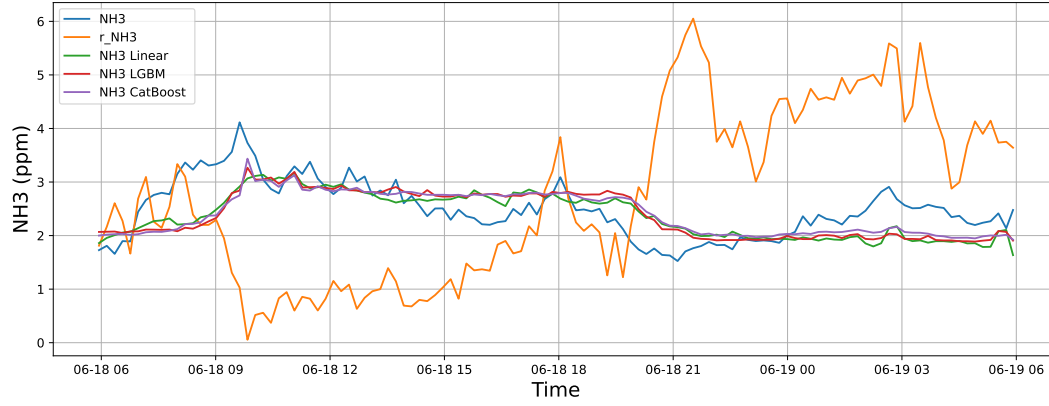


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

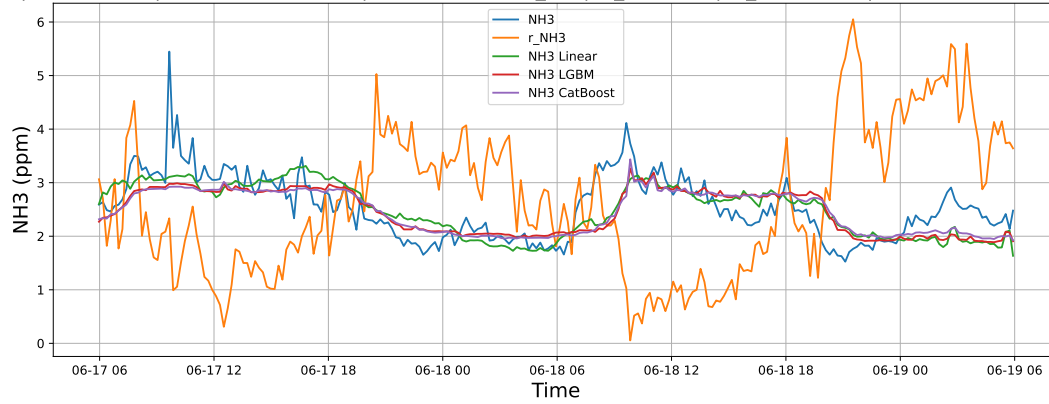
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



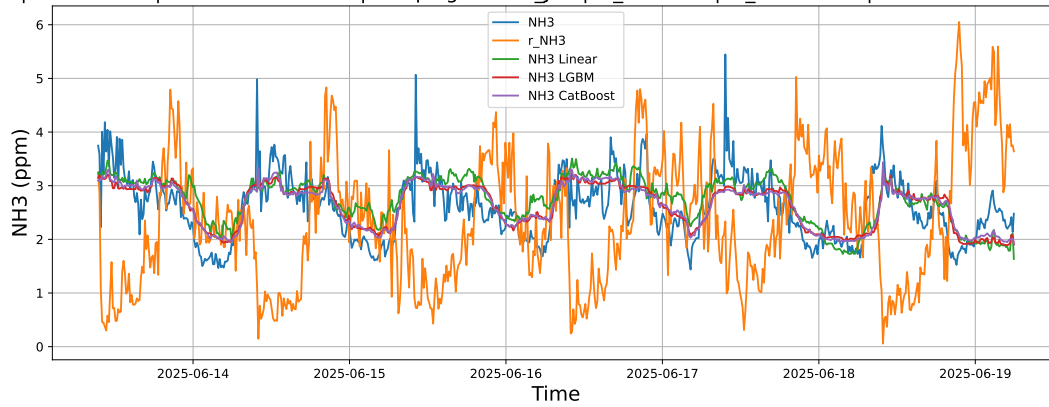
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

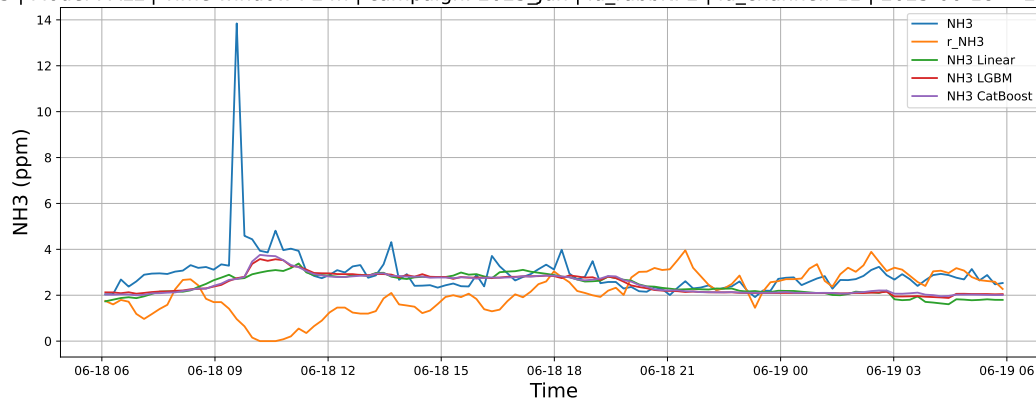
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

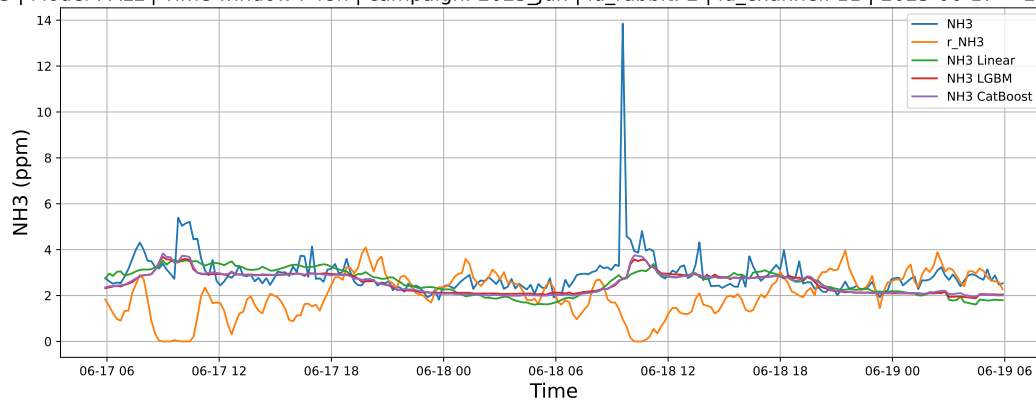
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



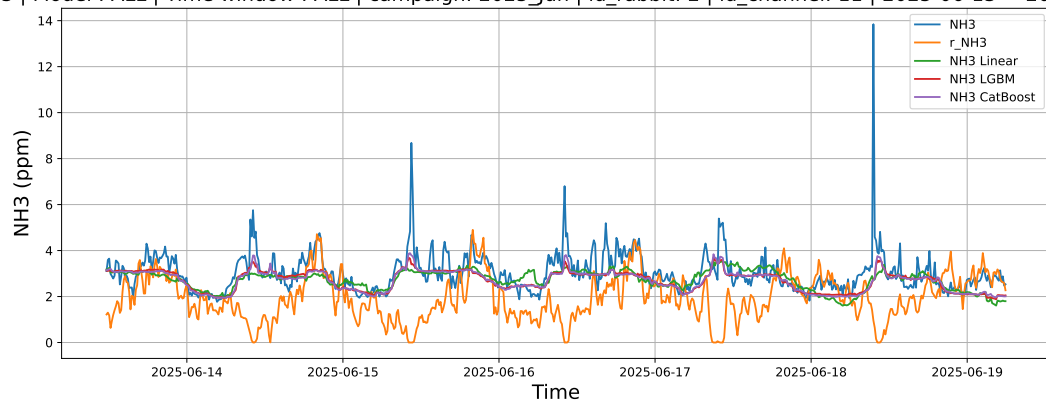
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

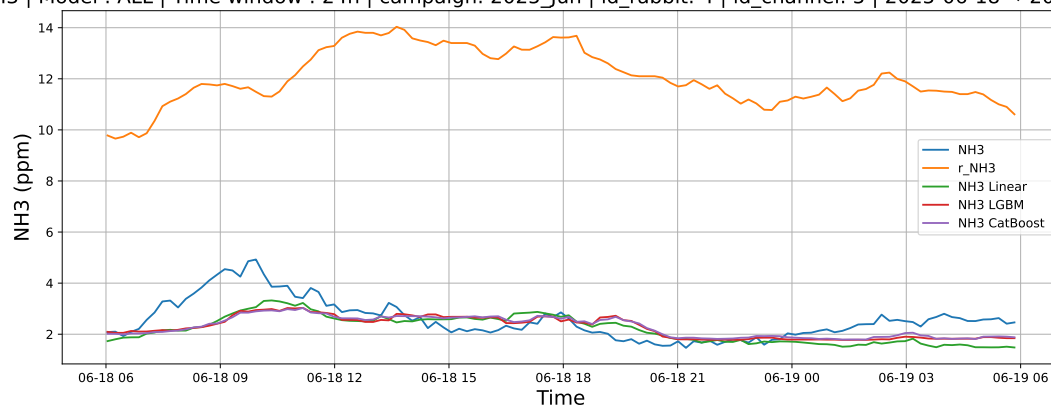
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

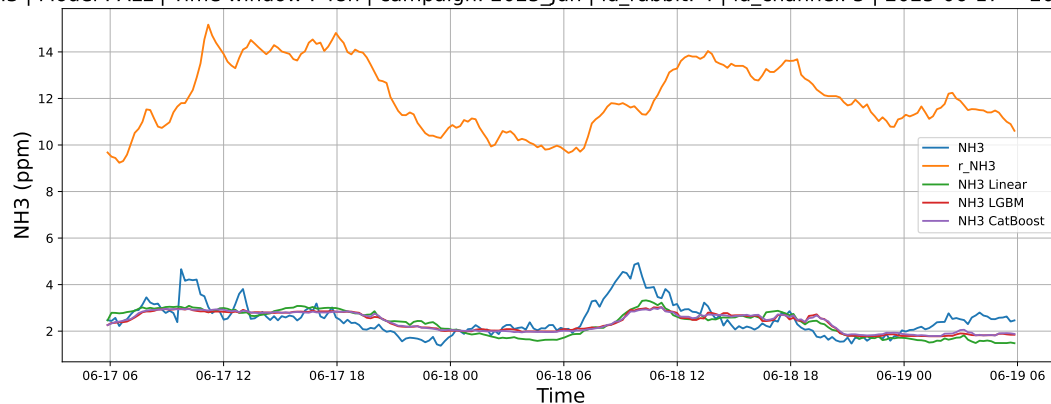
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



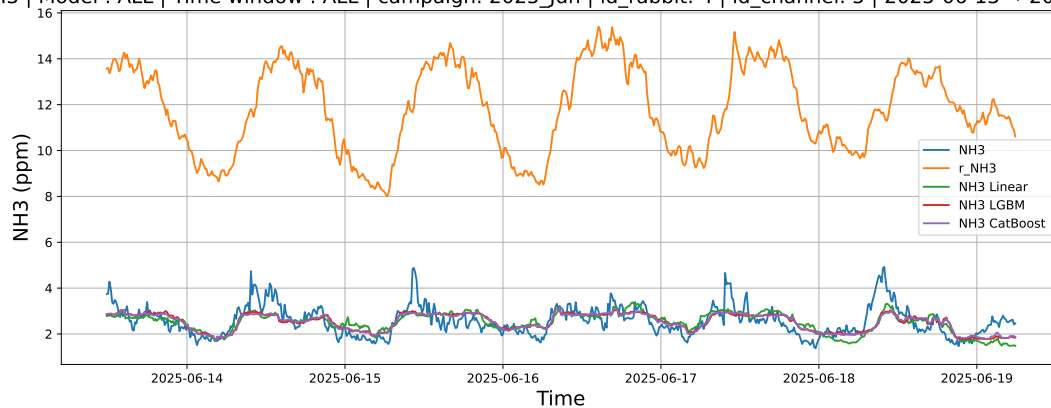
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

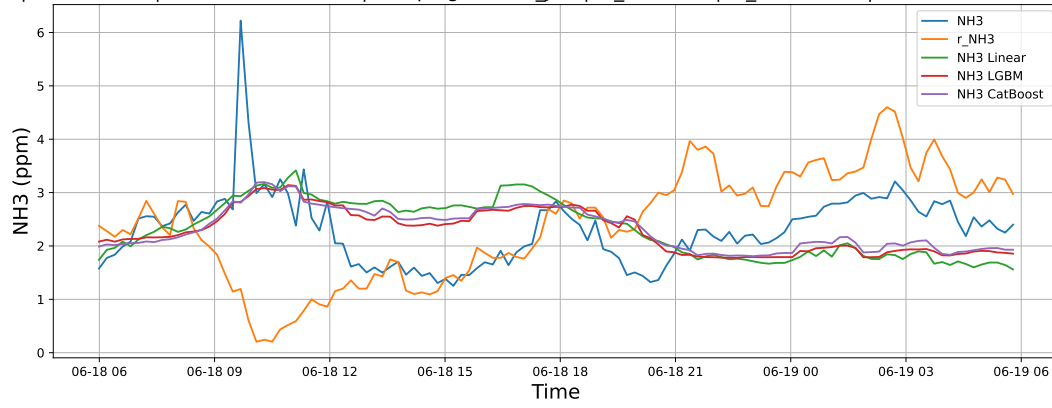
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

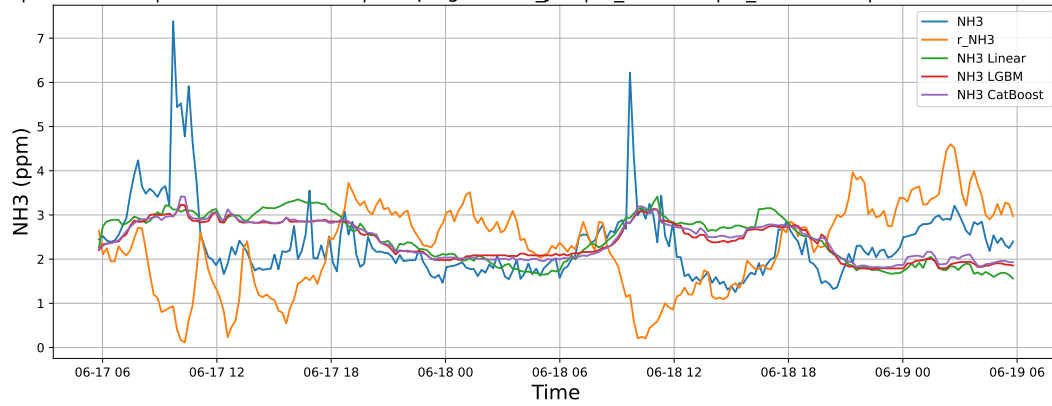
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



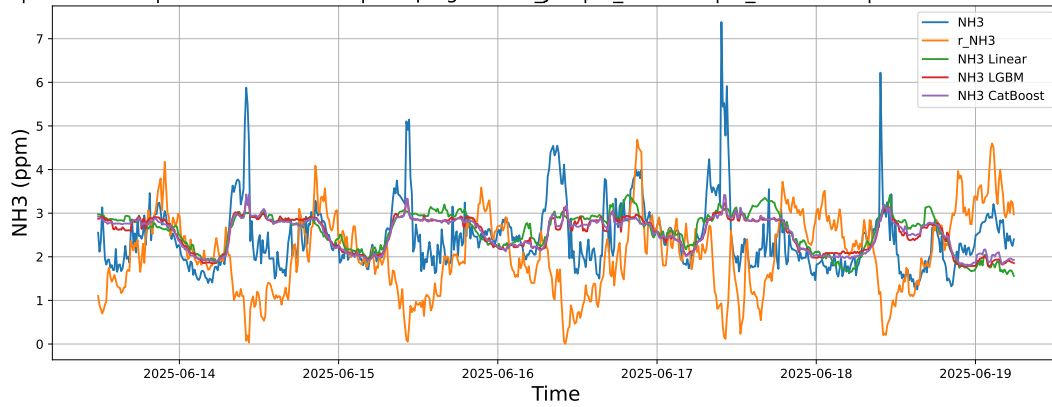
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

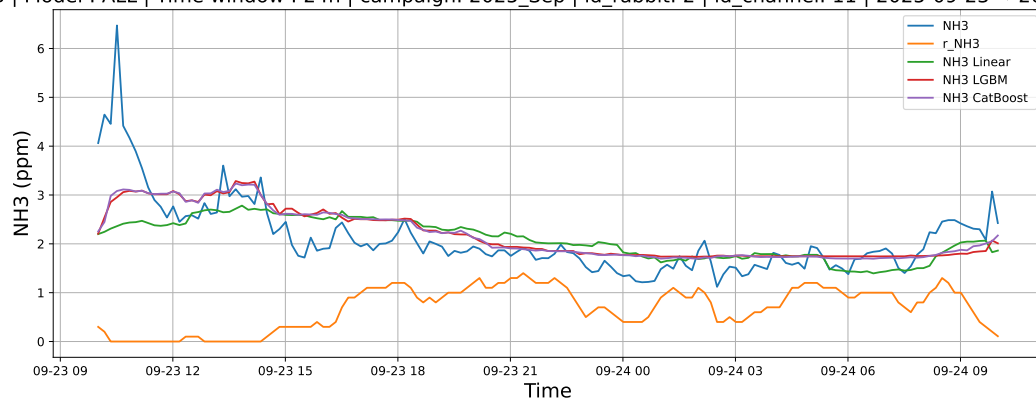
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

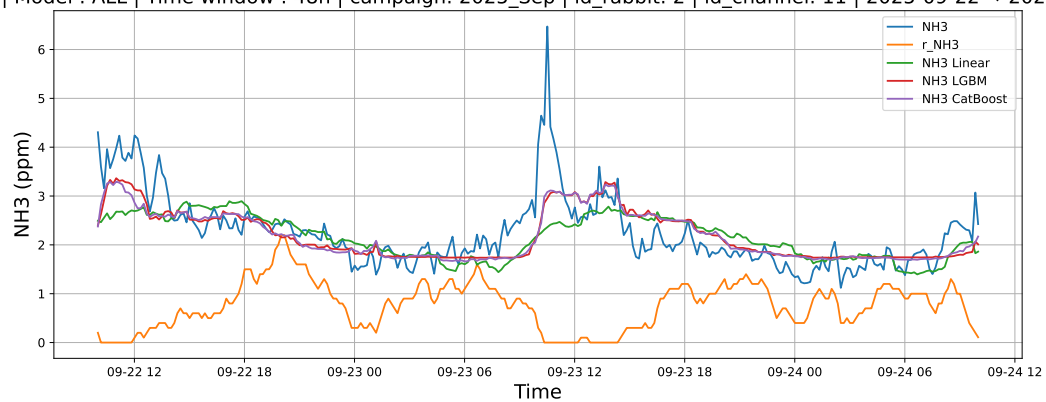
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



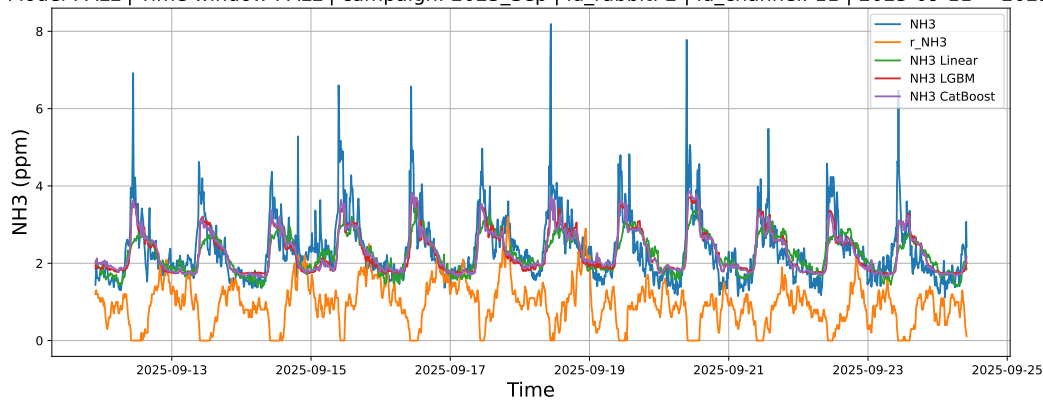
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

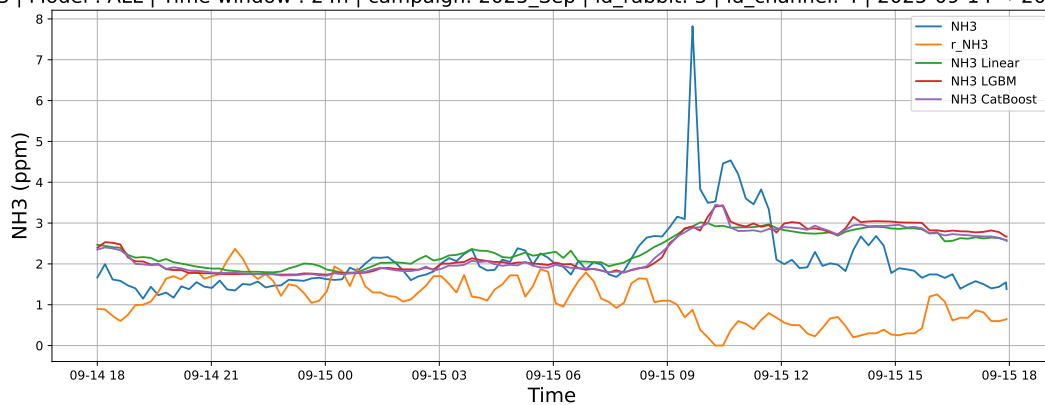
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

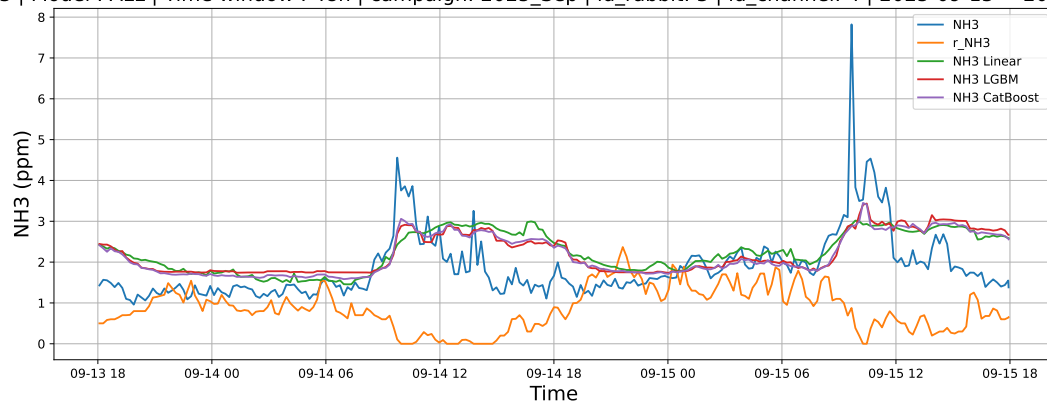
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



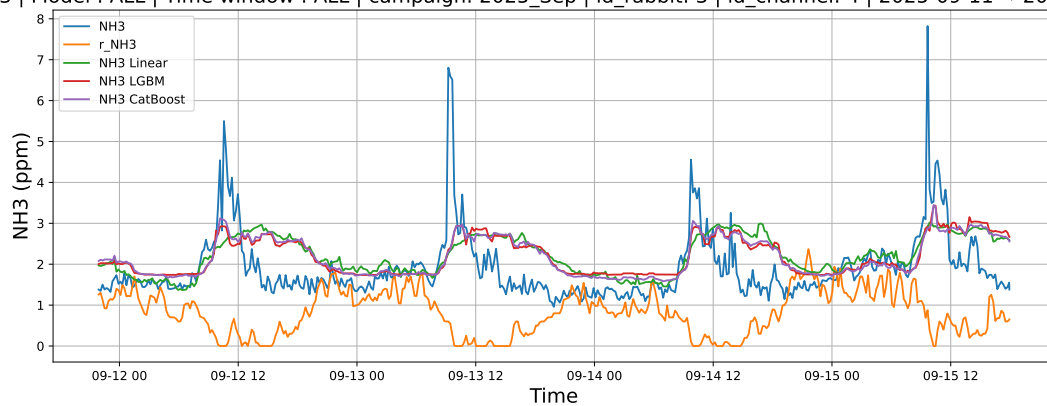
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

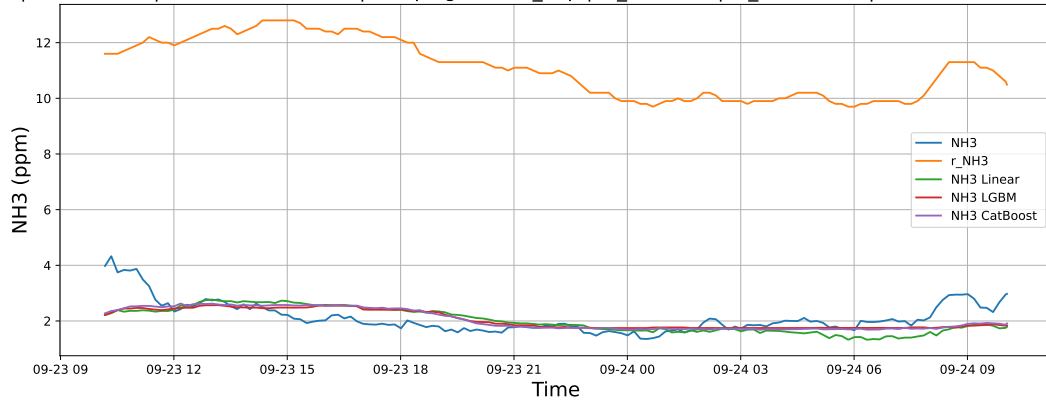
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

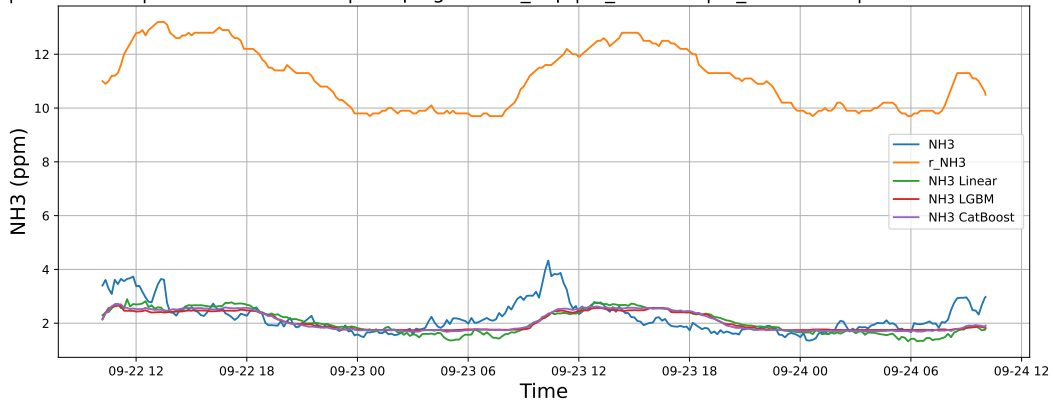
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



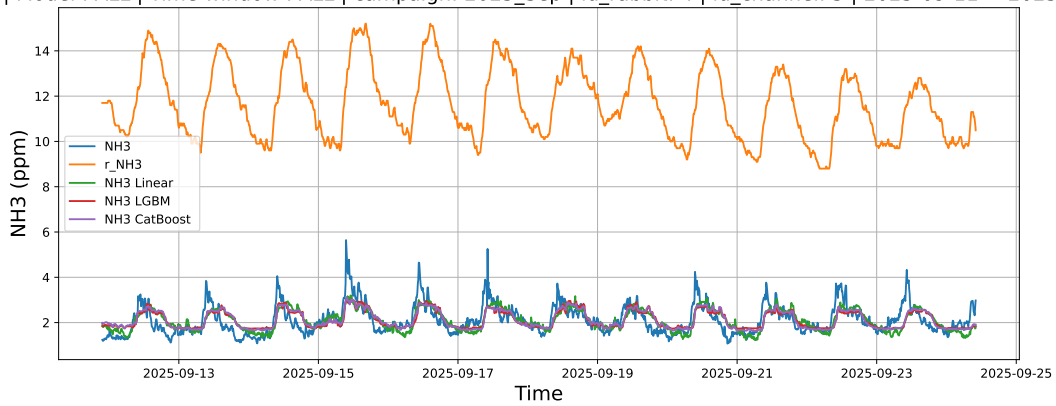
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

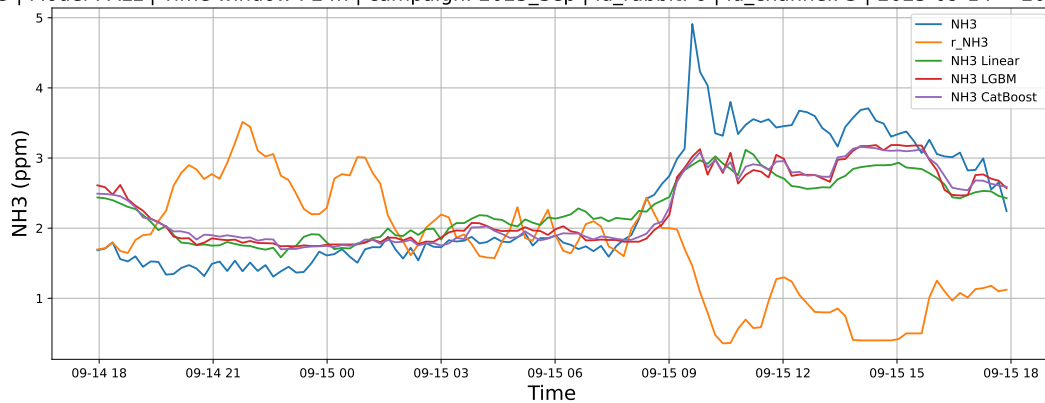
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

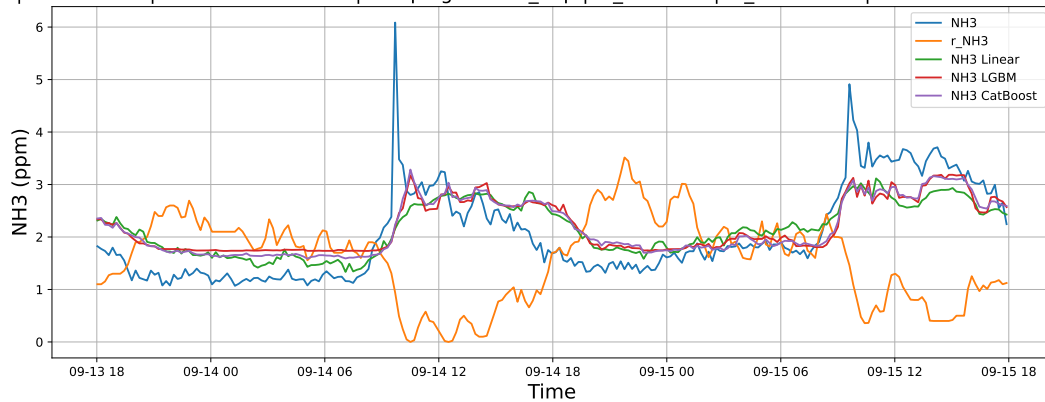
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15

